



The observability of fuzzy fractional evolution equations in the generalized linearly correlated fuzzy spaces

Nguyen Thi Thu Huyen^a, Nguyen Thi Kim Son^{b,*}, Hoang Thi Phuong Thao^c,
Nguyen Phuong Dong^d

^a Faculty of Mathematics and Information Technology, Hanoi Metropolitan University, Hanoi, Viet Nam

^b School of Information and Communications Technology, Hanoi University of Industry, Hanoi, Viet Nam

^c Foreign Language Specialized School, University of Languages and International Studies, Vietnam National University, Hanoi, Viet Nam

^d Faculty of Mathematics, Hanoi Pedagogical University 2, Phu Tho, Viet Nam

ARTICLE INFO

Keywords:

Strong linear independence
Fuzzy Fréchet derivative
Fuzzy Caputo fractional derivative
Observability

ABSTRACT

The work is devoted to studying the observability of fuzzy fractional evolution equations in the space of generalized linearly correlated fuzzy numbers, namely the space $\mathbb{LC}(A_1, A_2)$. Firstly, we develop an analysis of functions taking values in $\mathbb{LC}(A_1, A_2)$. Next, we present some conditions to ensure that the Cauchy problem for the proposed fuzzy fractional evolution equations admits a unique integral solution. Furthermore, by constructing an appropriate observability Gramian matrix, we obtain a sufficient condition such that the proposed fuzzy fractional evolution equations are observable. The concepts of derivative, fractional derivative and integral of the linearly correlated fuzzy functions in the space $\mathbb{LC}(A_1, A_2)$ are introduced via the construction of the mapping $\psi : \mathbb{R}^2 \rightarrow \mathbb{LC}(A_1, A_2)$. The necessary and sufficient conditions for the mapping ψ to be an isomorphism are also mentioned.

1. Introduction

Differential calculus for fuzzy-valued functions is based on the difference of fuzzy sets, which is defined with the help of the Hukuhara difference of level sets of fuzzy sets. This calculus was initially developed by Puri and Ralescu [27], Goetschel and Voxman [19], Seikkala [30], Kaleva [20] and Bede and Gal [8]. The notion of the Hukuhara difference together with an adequate differential calculus for interval-valued functions was generalized by Stefanini [37] and there were some detailed studies on this topic carried out by Stefanini and Bede [38], Chalco-Cano et al. [9] and Lupulescu [23,24]. Generalized differential calculus for fuzzy-valued functions was initiated by Bede and Gal [8]. Subsequently, differential equations involving Hukuhara derivatives have been extensively studied. Based on the Hukuhara difference, Galanis [18] developed differential and integral calculus within a semilinear space. In recent decades, fractional calculus and fractional differential equations have received growing attention due to their wide-ranging applications in mathematics and the applied sciences [1,2,4,5,14,33,34]. These theories have been applied in the modeling of various physical, chemical, and engineering processes. Two of the most influential works in the field of fractional calculus are the monograph by Podlubny [28] and that by Kilbas et al. [21]. Agarwal et al. [1] proposed the concept of solutions to fractional differential equations with uncertainty. They considered the concept of the fuzzy Riemann-Liouville fractional derivative based on Hukuhara

* Corresponding author.

E-mail address: sonntk@haui.edu.vn (N.T.K. Son).

<https://doi.org/10.1016/j.fss.2025.109573>

Received 7 January 2024; Received in revised form 14 June 2025; Accepted 18 August 2025

differentiability to solve fuzzy fractional differential equations. Subsequently, Arshad and Lupulescu [6] established several results concerning the existence and uniqueness of solutions to fuzzy fractional differential equations under Hukuhara differentiability. Alahviranloo et al. [2,3] studied the concepts of generalized Hukuhara fractional Riemann-Liouville and Caputo derivatives of fuzzy functions. In several other works, Salahshour et al. [29] obtained solutions to fuzzy fractional differential equations using the fuzzy Laplace transform method.

Observability is one of the most important concepts in control theory, playing essential roles in both theoretical and practical aspects with various applications in science and engineering, such as pole placement, structural decomposition, stabilizability and observer design. The observability of a system is defined as a structural property that allows one to infer the initial state of the given system from observations of its input-output behavior. From a theoretical research perspective, there have been many systematic and detailed studies on controllability and observability problems for dynamical systems described by differential equations and partial differential equations, see [13,14,36,39]. During the rapid development of fuzzy mathematics and fuzzy dynamical systems, studies on the controllability and observability of fuzzy differential systems and fuzzy partial differential equations have also achieved several noticeable results (see [7,11,12]). In addition, issues regarding the criteria for controllability, observability, or observer design in fuzzy fractional dynamical control systems have also attracted considerable attention (see [5,10,13,34]).

Recall that for a given fuzzy number A , by using the linear dependence, Esmi et al. [17] constructed a novel set $\mathbb{R}_{F(A)}$ which is called the set of linearly correlated fuzzy numbers. For each fuzzy number A , we can consider a function $\psi_A : \mathbb{R}^2 \rightarrow \mathbb{R}_{F(A)}$, given by $\psi_A(q, r) = qA \oplus r$. Accordingly, the space $\mathbb{R}_{F(A)}$ is constructed by combining a given fixed fuzzy number A with the element 1. From this, we can see that the space $\mathbb{LC}(A_1, A_2)$, formed by $qA_1 \oplus rA_2$, is an extension of the space $\mathbb{R}_{F(A)}$ when we replace 1 with any fuzzy number. The space $\mathbb{LC}(A_1, A_2)$ is more flexible because it is constructed from a linear combination of any two fuzzy numbers: $\{A_1, A_2\}$. For example, we consider a fuzzy number $A_1 = (0; 1; 2)$. We can see that the fuzzy number $u = (2; 6; 10) \notin \mathbb{R}_{F(A_1)}$ but $u \in \mathbb{LC}(A_1, A_2)$ with $A_1 = (0; 1; 2)$, $A_2 = (2; 3; 4)$ since $u = 3A_1 \oplus A_2$. In [17], the authors showed that $\mathbb{R}_{F(A)}$ becomes a Banach space when A is a non-symmetric fuzzy number. This leads to the condition under which the pair $\{A, 1\}$ becomes a strongly linearly independent (SLI) set. Similar to the space $\mathbb{R}_{F(A)}$, the structure of the space $\mathbb{LC}(A_1, A_2)$ is also affected by the SLI property of the set $\{A_1, A_2\}$. Specifically, if $\{A_1, A_2\}$ is SLI, the space $\mathbb{LC}(A_1, A_2)$ is a Banach space. Esmi et al. [17] considered the Fréchet derivative of linearly correlated fuzzy number-valued functions. For a fuzzy number A , the authors considered the space $\mathbb{R}_{F(A)} = \{qA \oplus r, (q, r) \in \mathbb{R}^2\}$. It is important to note that the structure of the space of linearly correlated fuzzy numbers depends on the symmetry of the considered fuzzy number. In [25], the authors developed a calculus for linearly correlated fuzzy processes via the Fréchet derivative and defined the concept of the Riemann integral. Later, in [32], Shen established both differential and integral calculus for linearly correlated non-symmetric fuzzy-valued functions. This study proposed a more general definition of derivatives (leading to Fréchet derivatives), extended these results to the case of symmetric fuzzy numbers, and discussed related properties. Recently, Shen [31] introduced two concepts of LC-differentiability and Fréchet differentiability in the space $\mathbb{R}_{F(A)}$ and applied them to study the first-order linear fuzzy differential equations. Moreover, the method for solving fuzzy differential systems taking values in the space of linearly correlated fuzzy numbers has also attracted considerable attention. For example, Pedro et al. [26] introduced the fuzzy Laplace transform for linearly correlated fuzzy-valued functions and applied it to solve several classes of fuzzy differential equations. Along with the study of integer-order derivatives and integrals in the space $\mathbb{R}_{F(A)}$, the theory of fractional calculus for such spaces has also gradually attracted considerable attention. In [35], the authors were the first to introduce the definitions of Fréchet Caputo fractional derivative, Fréchet Riemann-Liouville fractional derivative and Fréchet Caputo-Fabrizio fractional derivative for linearly correlated fuzzy-valued functions and applied these definitions to investigate several classes of fuzzy fractional differential systems. After that, in the work [33], Son studied the existence and uniqueness of fuzzy solutions to a nonlocal problem involving a multi-order fractional differential system whose state takes values in the space of linearly correlated fuzzy numbers.

Motivated by the work in [17], this paper considers the space $\mathbb{LC}(A_1, A_2)$ with the linear representation on A_1, A_2 , namely $\mathbb{LC}(A_1, A_2) = \{qA_1 \oplus rA_2 : (q, r) \in \mathbb{R}^2\}$. Note that in [17], the authors proved that $\mathbb{R}_{F(A)}$ is a Banach space if A is a non-symmetric fuzzy number. The question is whether there are any conditions on A_1, A_2 under which the space $\mathbb{LC}(A_1, A_2)$ is also a Banach space. The answer will be presented in the first part of the work. This forms an important foundation for studying the calculus of generalized linearly correlated fuzzy-valued functions. In particular, we introduce definitions for the integral, derivative and fractional calculus of linearly correlated fuzzy functions in $\mathbb{LC}(A_1, A_2)$ and establish several related properties. In addition, we study the space $\mathbb{LC}(A_1, A_2)$ in both cases, whether the set $\{A_1, A_2\}$ is SLI or not. Furthermore, one of the most interesting results of this work concerns the observability of fuzzy fractional evolution systems taking values in the space $\mathbb{LC}(A_1, A_2)$. The main contributions of this work are summarized as follows:

- 1) The first contribution of this work is the construction of the space of linearly correlated fuzzy numbers $\mathbb{LC}(A_1, A_2)$ and the establishment of several related properties. Under the assumption that $\{A_1, A_2\}$ does not satisfy the SLI property, we show that the mapping $\psi^{-1}(\psi(q, r))$ is well-defined. In addition, we identify the possible cases for the set $\psi^{-1}(\psi(q, r))$.
- 2) For the SLI set $\{A_1, A_2\}$, we introduce the Fréchet derivative of fuzzy functions taking values in $\mathbb{LC}(A_1, A_2)$. Meanwhile, we also formally extend this differentiability to the non-SLI set. Subsequently, we establish a fuzzy calculus framework for linearly correlated fuzzy number-valued functions in the space $\mathbb{LC}(A_1, A_2)$ under Fréchet differentiability. Several new concepts of the Fréchet derivative are introduced for both the integer-order and fractional-order cases.
- 3) Finally, we study the observability problem of fuzzy fractional evolution equations taking values in $\mathbb{LC}(A_1, A_2)^n$. In the case where the pair $\{A_1, A_2\}$ is SLI, by applying the Laplace transform, we provide the formula for the existence of a fuzzy solution to the Cauchy problem for the given fuzzy fractional evolution equations. Furthermore, we also give the formula for the observability Gramian matrix. From this result, we prove that the given fuzzy fractional evolution equations are observable.

2. Strong linear independence

In order to study dynamics in linearly correlated fuzzy number spaces, the linearity of the space $\mathbb{LC}(A_1, A_2)$ plays a key role. If the set $\{A_1, A_2\}$ is strong linearly independent (SLI), the map ψ becomes bijective, allowing for the transfer of structure and operations (such as addition and scalar multiplication) from $\mathbb{R}^2$ to $\mathbb{LC}(A_1, A_2)$. This enables proofs related to dynamical systems. However, if $\{A_1, A_2\}$ is not SLI, then the mapping ψ is not bijective and hence, this leads to extreme difficulty in constructing operations and determining dynamical properties within $\mathbb{LC}(A_1, A_2)$. To overcome this challenge, our initial step is to analyze the structure of $\mathbb{LC}(A_1, A_2)$ corresponding to different sets $\{A_1, A_2\}$. Next, we will examine the non-SLI cases of $\{A_1, A_2\}$, focusing on finding the inverse image of ψ to establish suitable operations for $\mathbb{LC}(A_1, A_2)$ in each case.

We recall that the space of all fuzzy numbers on $\mathbb{R}$ is denoted by the symbol $\mathbb{R}_F$. For each $A \in \mathbb{R}_F$, the α -cuts of A are given by

$$[A]^\alpha = \begin{cases} \{x \in \mathbb{R} : A(x) \geq \alpha\} & \text{if } \alpha \in (0, 1], \\ \{x \in \mathbb{R} : A(x) > 0\} & \text{if } \alpha = 0. \end{cases}$$

It is well-known that the α -cuts of A can be presented in the parametric form $[A]^\alpha = [A_\alpha^-, A_\alpha^+]$, $\alpha \in [0, 1]$. We denote $\text{len}[A](\alpha) = A_\alpha^+ - A_\alpha^-$, $\rho_A(\alpha) = A_\alpha^+ + A_\alpha^-$ for all $\alpha \in [0, 1]$. We recall that a fuzzy number $A \in \mathbb{R}_F$ is symmetric with respect to $x \in \mathbb{R}$ if $A(x - y) = A(x + y)$ for all $y \in \mathbb{R}$ and it is denoted by the symbol $(A|x)$. We say that A is non-symmetric if there does not exist any x such that A is symmetric. Let $n\mathbb{R}_F$ be the space of non-symmetric fuzzy numbers and $s\mathbb{R}_F$ be the space of symmetric fuzzy numbers except for real numbers.

From [15], let $\mathbb{LC}(A_1, A_2)$ be the subset of $\mathbb{R}_F$ generated by $\{A_1, A_2\}$ with $A_1, A_2 \in \mathbb{R}_F$ that is defined as follows:

$$\mathbb{LC}(A_1, A_2) = \{qA_1 \oplus rA_2 | (q, r) \in \mathbb{R}^2\}.$$

Since $0A_1 \oplus 0A_2 = \hat{0}$, we have that $\hat{0} \in \mathbb{LC}(A_1, A_2)$.

Next, we reintroduce the concept of strong linear independence for fuzzy sets in [15].

Definition 2.1 ([15], [16]). A set of fuzzy numbers $\{A_1, A_2\}$ is said to be strongly linearly independent (SLI) if the following implication holds for every $A \in \mathbb{LC}(A_1, A_2)$:

$$(A|0) \Rightarrow q = r = 0,$$

where $A = qA_1 \oplus rA_2$, $q, r \in \mathbb{R}$.

Consider the function $\psi : \mathbb{R}^2 \rightarrow \mathbb{LC}(A_1, A_2)$ given by

$$\psi(q, r) = qA_1 \oplus rA_2 \text{ with } q, r \in \mathbb{R}. \quad (1)$$

According to Theorem 1 in [15], if $\{A_1, A_2\}$ is SLI, then the function ψ is a bijection. Conversely, if the function ψ is a bijection from $\mathbb{R}^2$ to $\mathbb{LC}(A_1, A_2)$, then the set $\{A_1, A_2\}$ is strongly linearly independent.

From the results in [15], if $\{A_1, A_2\}$ is SLI, then the arithmetic operations on the space $\mathbb{LC}(A_1, A_2)$ such as addition, subtraction, and scalar product are well-defined as follows. For all $B = \psi(q_B, r_B)$, $C = \psi(q_C, r_C) \in \mathbb{LC}(A_1, A_2)$ and $\lambda \in \mathbb{R}$, we have

- (i) $B \oplus C = \psi(q_B + q_C, r_B + r_C)$,
- (ii) $\lambda \cdot B = \psi(\lambda q_B, \lambda r_B)$,
- (iii) $B \ominus C = \psi(q_B - q_C, r_B - r_C)$.

The metric on the space $\mathbb{LC}(A_1, A_2)$ is defined by

$$d(B, C) = |q_B - q_C| + |r_B - r_C| \text{ for all } B, C \in \mathbb{LC}(A_1, A_2).$$

Moreover, the space $\mathbb{LC}(A_1, A_2)$ endowed with metric d is a complete metric space. In this paper, $\mathbb{LC}(A_1, A_2)$ is called the generalized linearly correlated fuzzy space. In addition, if (A_1, A_2) is SLI, then the space $\mathbb{LC}(A_1, A_2)$ is isometric to the Banach space $\mathbb{R}^2$ and hence, the space $(\mathbb{LC}(A_1, A_2), \oplus, \cdot, \|\cdot\|)$ is a Banach space with the induced norm $\|B\| = \|\psi^{-1}(B)\|_{\mathbb{R}^2} = |q_B| + |r_B| = d(B, \hat{0})$ where $B = \psi(q_B, r_B) \in \mathbb{LC}(A_1, A_2)$ and $\hat{0} = \psi(0, 0)$ is the zero element of the space $\mathbb{LC}(A_1, A_2)$.

Proposition 2.1. If $(A_1|x^*)$ with $x^* \neq 0$ and $A_2 \in n\mathbb{R}_F$ then the pair $\{A_1, A_2\}$ is SLI.

Proof. From Theorem 2 in [15], in order to prove the pair $\{A_1, A_2\}$ is SLI, we will prove that the equality $\lambda_1 \rho_{A_1}(\alpha) + \lambda_2 \rho_{A_2}(\alpha) = 0$ implies that $\lambda_1 = \lambda_2 = 0$. Indeed, we consider the equality

$$\lambda_1 \rho_{A_1}(\alpha) + \lambda_2 \rho_{A_2}(\alpha) = 0. \quad (2)$$

Since A_1 is symmetric with respect to x^* then it implies that $\rho_{A_1}(\alpha) = 2x^*$. By substituting the equality $\rho_{A_1}(\alpha) = 2x^*$ into the equation (2), we directly get

$$2\lambda_1 x^* + \lambda_2 A_{2\alpha}^+ + \lambda_2 A_{2\alpha}^- = 0.$$

If $\lambda_2 = 0$ then it implies that $2\lambda_1 x^* = 0$ and hence, it yields $\lambda_1 = 0$. In the case $\lambda_2 \neq 0$, we immediately get that

$$2\lambda_1 x^* + \lambda_2 (A_{2\alpha}^+ + A_{2\alpha}^-) = 0.$$

Let us choose $\frac{-\lambda_1}{\lambda_2} x^* = y^*$. Then, we immediately get that $A_{2\alpha}^+ + A_{2\alpha}^- = 2y^*$, or equivalently, $(A_2|y^*)$, which contradicts to our assumption. $\square$

Remark 2.1. Based on Esmi et al. [15] and Proposition 2.1, the SLI property of $\{A_1, A_2\}$ is concluded into the following cases, depending on the characteristics of A_1 and A_2 .

Case 1: A_1 and A_2 are non-symmetric. The SLI property depends on the linear independence of p_{A_1} and p_{A_2} (Theorem 2 in [15]) as follows:

1a) If $p_{A_2} = kp_{A_1}$, with $k \neq 0$, $\{A_1, A_2\}$ is not SLI.

1b) If $p_{A_2} \neq kp_{A_1}$, with $k \neq 0$, $\{A_1, A_2\}$ is SLI.

Case 2: A_1 is symmetric and A_2 is non-symmetric.

2a) If $(A_1|0)$, $A_2 \in n\mathbb{R}_F$, $\{A_1, A_2\}$ is not SLI (Corollary 3 in [15]).

2b) If $(A_1|x^*)$ with $x^* \neq 0$, $A_2 \in n\mathbb{R}_F$, $\{A_1, A_2\}$ is SLI (Proposition 2.1).

Case 3: If A_1 and A_2 are symmetric, $\{A_1, A_2\}$ is not SLI (Corollary 3 and Proposition 4 in [15]).

In order to study the dynamical properties in the space $\mathbb{LC}(A_1, A_2)$, we will explore the properties of the mapping ψ defined by (1) for each case in Remark 2.1. It is easy to see that in Case 1b and Case 2b, $\{A_1, A_2\}$ is SLI and ψ is a bijective mapping. Thus, in the following subsection, we will concretely consider the pre-image of ψ in the cases when $\{A_1, A_2\}$ is not SLI and ψ is not bijective. For Case 1a, the results are presented in Theorem 2.1 and Theorem 2.2. For Case 2a, the results are presented in Theorem 2.3 and in Case 3, the results are concretely presented in Theorems 2.4, 2.5 and 2.6.

Firstly, if $A_1, A_2 \in n\mathbb{R}_F$, then $\{A_1, A_2\}$ is not SLI when $\{\rho_{A_1}(\alpha), \rho_{A_2}(\alpha)\}$ is not linearly independent, which means that there exists $k \in \mathbb{R} \setminus \{0\}$ such that $\rho_{A_2}(\alpha) = k\rho_{A_1}(\alpha)$ for all $\alpha \in [0, 1]$. Now, we denote

$$K_1(\alpha) = \text{len}[A_2](\alpha) - k\text{len}[A_1](\alpha), K_2(\alpha) = \text{len}[A_2](\alpha) + k\text{len}[A_1](\alpha)$$

and

$$K_{i,j} = \frac{K_i(\alpha)}{K_j(\alpha)}, I_{i,j} = \frac{\text{len}[A_i](\alpha)}{K_j(\alpha)},$$

for all $i, j \in \{1, 2\}$, $\alpha \in [0, 1]$ and $k \in \mathbb{R} \setminus \{0\}$. Assume that there exists a pair $(u, v) \in \mathbb{R}^2$ such that $qA_1 \oplus rA_2 = uA_1 \oplus vA_2$ with $A_1, A_2 \in n\mathbb{R}_F$. We will find the relationship between (u, v) and (q, r) . For this aim, we divide the problem into two cases with respect to the signs of q, r through the following two theorems:

Theorem 2.1. Let $A_1, A_2 \in n\mathbb{R}_F$ and assume that there exists $k \in \mathbb{R} \setminus \{0\}$ such that $\rho_{A_2}(\alpha) = k\rho_{A_1}(\alpha)$ for all $\alpha \in [0, 1]$. If $(q, r) \in \mathbb{R}^2$, $A = qA_1 \oplus rA_2$ and q, r have the same sign then the inverse image of $\psi(q, r)$ is given by $\psi^{-1}(\psi(q, r))$ as follows:

(1) If (B1) or (B2) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (qK_{1,2}, 2ql_{1,2} + r)\}.$$

(2) If (B3) or (B4) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (q + 2krl_{2,2}, -K_{1,2}r)\}.$$

(3) If (B5) or (B6) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (K_{2,1}q + 2krl_{2,1}, -2ql_{1,1} - K_{2,1}r)\},$$

where the conditions $(B_i)_{i=1,6}$ are shown in Table 1.

Proof. Since $qA_1 \oplus rA_2 = uA_1 \oplus vA_2$ then it implies that u, v always satisfy the equation

$$u\rho_{A_1}(\alpha) + v\rho_{A_2}(\alpha) = q\rho_{A_1}(\alpha) + r\rho_{A_2}(\alpha)$$

According to the assumption $\rho_{A_2}(\alpha) = k\rho_{A_1}(\alpha)$, we directly get that $u + kv = q + kr$. Since $qr > 0$ then we receive the following cases:

Table 1Conditions for the representation of $\psi^{-1}(\psi(q, r))$ in Theorem 2.1.

Conditions	Description
(B1)	$k > 0, K_1(\alpha) \leq 0$ for all $\alpha \in [0, 1]$
(B2)	$k < 0, K_2(\alpha) < 0, r \geq -2ql_{1,2}$ if $r > 0$ and $r < -2ql_{1,2}$ if $r < 0$ for all $\alpha \in [0, 1]$
(B3)	$k > 0, K_1(\alpha) > 0$ for all $\alpha \in [0, 1]$
(B4)	$k < 0, K_2(\alpha) > 0, r \leq \frac{-q}{2kl_{2,2}}$ if $r > 0$ and $r > \frac{-q}{2kl_{2,2}}$ if $r < 0$ for all $\alpha \in [0, 1]$
(B5)	$k < 0, K_2(\alpha) < 0, r < -2ql_{1,2}$ if $r > 0$ and $r \geq -2ql_{1,2}$ if $r < 0$ for all $\alpha \in [0, 1]$
(B6)	$k < 0, K_2(\alpha) > 0, r > \frac{-q}{2kl_{2,2}}$ if $r > 0$ and $r \leq \frac{-q}{2kl_{2,2}}$ if $r < 0$ for all $\alpha \in [0, 1]$

Case 1: If $q > 0, r > 0, u \geq 0, v \geq 0$ or $q < 0, r < 0, u < 0, v < 0$, then u, v satisfy the system

$$\begin{cases} u\text{len}[A_1](\alpha) + v\text{len}[A_2](\alpha) = q\text{len}[A_1](\alpha) + r\text{len}[A_2](\alpha), \text{ for all } \alpha \in [0, 1] \\ u + kv = q + kr. \end{cases}$$

This implies

$$\begin{cases} (u - q)\text{len}[A_1](\alpha) + (v - r)\text{len}[A_2](\alpha) = 0 \\ u - q = k(r - v). \end{cases}$$

From this system, we have

$$k(r - v)\text{len}[A_1](\alpha) + (v - r)\text{len}[A_2](\alpha) = 0 \Rightarrow (r - v)[k\text{len}[A_1](\alpha) - \text{len}[A_2](\alpha)] = 0,$$

which means that $r = v$ and $u = q$.

Case 2: If $q > 0, r > 0, u < 0, v \geq 0$ or $q < 0, r < 0, u \geq 0, v < 0$, then u, v satisfy the system

$$\begin{cases} -u\text{len}[A_1](\alpha) + v\text{len}[A_2](\alpha) = q\text{len}[A_1](\alpha) + r\text{len}[A_2](\alpha) \\ u + kv = q + kr, \end{cases}$$

for all $\alpha \in [0, 1]$. Then, by solving the above system, we receive $u = K_{1,2}q$ and $v = 2l_{1,2}q + r$. Now, we will consider the sign conditions of u, v, q, r in each case. Note that $qu \leq 0$ and hence, it follows that $K_{1,2} \leq 0$.

For $k > 0$, we have $\text{len}[A_2](\alpha) + k\text{len}[A_1](\alpha) > 0$ and hence, $K_2(\alpha) > 0$. As a result, if u, v, q, r satisfy both cases, then $K_1(\alpha) < 0$ for all $\alpha \in [0, 1]$. In the case $k < 0$, it is true that $\text{len}[A_2](\alpha) - k\text{len}[A_1](\alpha) > 0$ and hence, we have $K_1(\alpha) > 0$. In case where $q > 0, r > 0, u < 0, v > 0$, in order to get $v > 0$, we need $r > -2ql_{1,2}$. In case $q < 0, r < 0, u > 0, v < 0$, in order to get $v < 0$, we need $r < -2ql_{1,2}$. Therefore, if q, r satisfy the condition (B2), then we have $u = K_{1,2}q$ and $v = 2l_{1,2}q + r$. Thus, if $qr > 0$ and the condition (B1) or (B2) is satisfied then $\psi^{-1}(\psi(q, r)) = \{(q, r), (qK_{1,2}, 2ql_{1,2} + r)\}$.

Finally, by doing similar arguments, we receive the results for the remaining cases:

Case 3: $q > 0, r > 0, u \geq 0, v < 0$ or $q < 0, r < 0, u < 0, v \geq 0$.

Case 4: If $q > 0, r > 0, u < 0, v < 0$ or $q < 0, r < 0, u \geq 0, v \geq 0$.

The proof is completed. $\square$

By applying similar arguments to those in Theorem 2.1, we obtain the following result for the case where q and r have opposite signs.

Theorem 2.2. Let $A_1, A_2 \in n\mathbb{R}_F$ and assume that there exists $k \in \mathbb{R} \setminus \{0\}$ such that $\rho_{A_2}(\alpha) = k\rho_{A_1}(\alpha)$ for all $\alpha \in [0, 1]$. For $(q, r) \in \mathbb{R}^2$, let $A = qA_1 \oplus rA_2$ where q, r have opposite signs. Then, the inverse image of $\psi(q, r)$ is given by $\psi^{-1}(\psi(q, r))$ as follows: If (B7) or (B8) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (K_{2,1}q, -2ql_{1,1} + r)\}.$$

If (B9) or (B10) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (q + 2krl_{2,1}, -K_{2,1}r)\}.$$

If (B11) or (B12) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (K_{1,2}q + 2krl_{2,2}, 2ql_{1,2} - K_{1,2}r)\},$$

where the conditions $(B_i)_{i=7,12}$ are shown in Table 2.

Table 2
Conditions for the representation of $\psi^{-1}(\psi(q, r))$ in Theorem 2.2.

Conditions	Description
(B7)	$k < 0, K_2(\alpha) < 0$ for all $\alpha \in [0, 1]$
(B8)	$k > 0, K_1(\alpha) < 0, r \geq 2ql_{1,1}$ if $r > 0$ and $r < 2ql_{1,1}$ if $r < 0$ for all $\alpha \in [0, 1]$
(B9)	$k < 0, K_2(\alpha) > 0$ for all $\alpha \in [0, 1]$
(B10)	$k > 0, K_1(\alpha) > 0, r < \frac{-q}{2kl_{2,1}}$ if $r > 0$ and $r \geq \frac{-q}{2kl_{2,1}}$ if $r < 0$ for all $\alpha \in [0, 1]$
(B11)	$k > 0, K_1(\alpha) < 0, r < 2ql_{1,1}$ if $r > 0$ and $r \geq 2ql_{1,1}$ if $r < 0$ for all $\alpha \in [0, 1]$
(B12)	$k > 0, K_1(\alpha) > 0, r \geq \frac{-q}{2kl_{2,1}}$ if $r > 0$ and $r < \frac{-q}{2kl_{2,1}}$ if $r < 0$ for all $\alpha \in [0, 1]$

Example 2.1. Consider $A_1 = \left(\frac{-1}{2}; 0; 1\right)$ and $A_2 = \left(\frac{-1}{2}; 0; 0\right)$ to be non-symmetric fuzzy numbers. For the fuzzy number $A_1 = \left(\frac{-1}{2}; 0; 1\right)$, we have

$$\begin{aligned} A_{1\alpha}^- &= \frac{-1}{2} + \frac{1}{2}\alpha, \\ A_{1\alpha}^+ &= 1 - \alpha, \\ \text{len}[A_1](\alpha) &= \frac{3}{2} - \frac{3}{2}\alpha, \\ \rho_{A_1}(\alpha) &= \frac{1}{2} - \frac{1}{2}\alpha, \end{aligned}$$

for all $\alpha \in [0, 1]$. Next, for the fuzzy number $A_2 = \left(\frac{-1}{2}; 0; 0\right)$, we have

$$\begin{aligned} A_{2\alpha}^- &= \frac{-1}{2} + \frac{1}{2}\alpha, \\ A_{2\alpha}^+ &= 0, \\ \text{len}[A_2](\alpha) &= \frac{1}{2} - \frac{1}{2}\alpha, \\ \rho_{A_2}(\alpha) &= \frac{1}{2} - \frac{1}{2}\alpha, \end{aligned}$$

for all $\alpha \in [0, 1]$. As a result, we can see that $\rho_{A_2}(\alpha) = -\rho_{A_1}(\alpha)$ for all $\alpha \in [0, 1]$. According to Theorem 2.1, it implies that

$$k = -1 < 0, K_2(\alpha) = \text{len}[A_2](\alpha) + (-1)\text{len}[A_1](\alpha) = -1 + \alpha \leq 0 \text{ for all } \alpha \in [0, 1].$$

Thus, it follows that

$$\begin{aligned} l_{1,2} &= \frac{\text{len}[A_1](\alpha)}{K_2(\alpha)} = \frac{-3}{2}, \\ l_{2,2} &= \frac{\text{len}[A_2](\alpha)}{K_2(\alpha)} = \frac{-1}{2}. \end{aligned}$$

Let $A \in \mathbb{LC}(A_1, A_2)$ with $A = A_1 \oplus A_2$, that is, $q = 1, r = 1$. Then, we have $-2ql_{1,2} = 3$ and $r < -2ql_{1,2}$. From Theorem 2.1, we have

$$u = K_{2,1}q + 2krl_{2,1} = -1; \quad v = -2ql_{1,1} - K_{2,1}r = -1$$

Therefore, $\psi^{-1}(\psi(1, 1)) = \{(1, 1), (-1; -1)\}$.

Theorem 2.3. If $(A_1|0)$, $A_2 \in n\mathbb{R}_F$ and $A = qA_1 \oplus rA_2$ with $q, r \in \mathbb{R}^2$ then the inverse image of the function $\psi(q, r)$ is given by

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (-q, r)\}.$$

Proof. Assume that there exist $u, v \in \mathbb{R}$ such that $qA_1 \oplus rA_2 = uA_1 \oplus vA_2$. For each $\alpha \in [0, 1]$, we have the following system

$$\begin{cases} uA_{1\alpha}^+ + vA_{2\alpha}^+ = qA_{1\alpha}^+ + rA_{2\alpha}^+, \\ uA_{1\alpha}^- + vA_{2\alpha}^- = qA_{1\alpha}^- + rA_{2\alpha}^-, \end{cases}$$

which means that u and v always satisfy the equation $u\rho_{A_1}(\alpha) + v\rho_{A_2}(\alpha) = q\rho_{A_1}(\alpha) + r\rho_{A_2}(\alpha)$. Since $(A_1|0)$, that is, $\rho_{A_1} = 0$, then we directly get that $v = r$. Now, we consider the following cases:

Case 1: If $qr \geq 0$, that is, $q \geq 0, r \geq 0, u \geq 0, v \geq 0$ or $q < 0, r < 0, u < 0, v < 0$ then u and v satisfy the following system

$$\begin{cases} u\text{len}[A_1](\alpha) + v\text{len}[A_2](\alpha) = q\text{len}[A_1](\alpha) + r\text{len}[A_2](\alpha), \\ u\rho_{A_1}(\alpha) + v\rho_{A_2}(\alpha) = q\rho_{A_1}(\alpha) + r\rho_{A_2}(\alpha), \end{cases}$$

for all $\alpha \in [0, 1]$. Next, by solving the above system, we receive $v = r$ and $u = q$.

If $q \geq 0, r \geq 0, u < 0, v \geq 0$ or $q < 0, r < 0, u \geq 0, v < 0$ then u, v satisfy the system

$$\begin{cases} -u\text{len}[A_1](\alpha) + v\text{len}[A_2](\alpha) = q\text{len}[A_1](\alpha) + r\text{len}[A_2](\alpha), \\ u\rho_{A_1}(\alpha) + v\rho_{A_2}(\alpha) = q\rho_{A_1}(\alpha) + r\rho_{A_2}(\alpha), \end{cases}$$

for all $\alpha \in [0, 1]$. This system yields $v = r, u = -q$.

In the remaining four cases $\{q \geq 0, r \geq 0, u \geq 0, v < 0\}$, $\{q < 0, r < 0, u < 0, v \geq 0\}$, $\{q \geq 0, r \geq 0, u < 0, v < 0\}$ and $\{q < 0, r < 0, u \geq 0, v \geq 0\}$, it is true that v and r are always of opposite signs and are non-zero at the same time. Thus, there do not exist q, r, u, v satisfying the requirements.

Case 2: $qr < 0$. Similar to Case 1, we get the result as follows:

If $q > 0, r < 0, u \geq 0, v < 0$ or $q < 0, r > 0, u < 0, v \geq 0$, we obtain $v = r$ and $u = q$.

If $q > 0, r < 0, u < 0, v < 0$ or $q < 0, r > 0, u \geq 0, v \geq 0$, we obtain $v = r$ and $u = -q$.

In the rest cases $\{q > 0, r < 0, u \geq 0, v \geq 0\}$, $\{q < 0, r > 0, u < 0, v < 0\}$, $\{q > 0, r < 0, u < 0, v \geq 0\}$ and $\{q < 0, r > 0, u \geq 0, v < 0\}$, we can see that v and r are always of opposite signs and hence, there does not exist q, r, u, v satisfying the requirements. $\square$

Example 2.2. Let $A_1 = (-1; 0; 1)$ be a symmetric fuzzy number with respect to $x^* = 0$, $A_2 = (1; 3; 4)$ and $A = (1; 6; 9)$ be non-symmetric fuzzy numbers. Here, we can verify that the fuzzy number A can be represented by $A = A_1 \oplus 2A_2$. Therefore, we can point out that $A \in \mathbb{LC}(A_1, A_2)$ with $q = 1$ and $r = 2$. According to Theorem 2.3, we have $\psi^{-1}(\psi(1; 2)) = \{(1; 2), (-1; 2)\}$. Indeed, we can see that $A = A_1 \oplus 2A_2 = (-1)A_1 \oplus 2A_2$.

Our next goal is to determine the image of the map ψ when A_1 and A_2 are symmetric fuzzy numbers. To simplify this, we divide the problem into two separate cases: $A_1|0, A_2|0$ and $A_1|x_1, A_2|x_2$ with $x_1 \neq 0$. We will start by considering the first case.

For $\alpha \in [0, 1]$ and $i, j \in \{1, 2\}$ set

$$\bar{k} = \frac{x_2}{x_1}, \bar{K}_1(\alpha) = \text{len}[A_2](\alpha) - \bar{k}\text{len}[A_1](\alpha), \bar{K}_2(\alpha) = \text{len}[A_2](\alpha) + \bar{k}\text{len}[A_1](\alpha)$$

and

$$\bar{K}_{i,j} = \frac{\bar{K}_i(\alpha)}{\bar{K}_j(\alpha)}, \bar{l}_{i,j} = \frac{\text{len}[A_i](\alpha)}{\bar{K}_j(\alpha)}.$$

Note that, if $(A_1|x_1)$ and $(A_2|x_2)$, then $\rho_{A_1}(\alpha) = 2x_1$ and $\rho_{A_2}(\alpha) = 2x_2$. Applying the same proof method as presented in Theorem 2.1, we obtain the following results.

Theorem 2.4. If $(A_1|x_1), (A_2|x_2)$ with $x_1 \neq 0$ and $A = qA_1 \oplus rA_2$ for $(q, r) \in \mathbb{R}^2$, where q, r have opposite signs then the inverse image of $\psi(q, r)$ is given by $\psi^{-1}(\psi(q, r))$ as follows:

If (C1) or (C2) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (\bar{K}_{2,1}q, -2q\bar{l}_{1,1} + r)\}.$$

If (C3) or (C4) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (q + 2\bar{k}r\bar{l}_{2,1}, -\bar{K}_{2,1}r)\}.$$

If (C5) or (C6) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (\bar{K}_{1,2}q + 2\bar{k}r\bar{l}_{2,2}, 2q\bar{l}_{1,2} - \bar{K}_{1,2}r)\},$$

where the conditions $(C_i)_{i=1,6}$ are shown in Table 3.

Theorem 2.5. If $(A_1|x_1)$ and $(A_2|x_2)$ with $x_1 \neq 0$ and $A = qA_1 \oplus rA_2$ for $(q, r) \in \mathbb{R}^2$, where q, r have the same sign then the inverse image of $\psi(q, r)$ is given by $\psi^{-1}(\psi(q, r))$ as follows:

If (C7) or (C8) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (q\bar{K}_{1,2}, 2q\bar{l}_{1,2} + r)\}.$$

If (C9) or (C10) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (q + 2\bar{k}r\bar{l}_{2,2}, -\bar{K}_{1,2}r)\}.$$

Table 3
Conditions for the representation of $\psi^{-1}(\psi(q, r))$ in Theorem 2.4.

Conditions	Description
(C1)	$\bar{k} < 0, \bar{K}_2(\alpha) \leq 0$ for all $\alpha \in [0, 1]$
(C2)	$\bar{k} > 0, \bar{K}_1(\alpha) < 0, r \geq 2q\bar{l}_{1,1}$ if $r > 0$ and $r < 2q\bar{l}_{1,1}$ if $r < 0$ for all $\alpha \in [0, 1]$
(C3)	$\bar{k} < 0, \bar{K}_2(\alpha) > 0$ for all $\alpha \in [0, 1]$
(C4)	$\bar{k} > 0, \bar{K}_1(\alpha) > 0, r < \frac{-q}{2kl_{2,1}}$ if $r > 0$ and $r \geq \frac{-q}{2kl_{2,1}}$ if $r < 0$ for all $\alpha \in [0, 1]$
(C5)	$\bar{k} > 0, \bar{K}_1(\alpha) < 0, r < 2q\bar{l}_{1,1}$ if $r > 0$ and $r \geq 2q\bar{l}_{1,2}$ if $r < 0$ for all $\alpha \in [0, 1]$
(C6)	$\bar{k} > 0, \bar{K}_1(\alpha) > 0, r \geq \frac{-q}{2kl_{2,1}}$ if $r > 0$ and $r < \frac{-q}{2kl_{2,1}}$ if $r < 0$ for all $\alpha \in [0, 1]$

Table 4
Conditions for the representation of $\psi^{-1}(\psi(q, r))$ in Theorem 2.5.

Conditions	Description
(C7)	$\bar{k} > 0, \bar{K}_1(\alpha) \leq 0$ for all $\alpha \in [0, 1]$
(C8)	$\bar{k} < 0, \bar{K}_2(\alpha) < 0, r \geq -2q\bar{l}_{1,2}$ if $r > 0$ and $r < -2q\bar{l}_{1,2}$ if $r < 0$ for all $\alpha \in [0, 1]$
(C9)	$\bar{k} > 0, \bar{K}_1(\alpha) > 0$ for all $\alpha \in [0, 1]$
(C10)	$\bar{k} < 0, \bar{K}_2(\alpha) > 0, r \geq \frac{-q}{2kl_{2,2}}$ if $r > 0$ and $r > \frac{-q}{2kl_{2,2}}$ if $r < 0$ for all $\alpha \in [0, 1]$
(C11)	$\bar{k} < 0, \bar{K}_2(\alpha) > 0, r > \frac{-q}{2kl_{2,2}}$ if $r > 0$ and $r \leq \frac{-q}{2kl_{2,2}}$ if $r < 0$ for all $\alpha \in [0, 1]$
(C12)	$\bar{k} < 0, \bar{K}_2(\alpha) < 0, r < -2q\bar{l}_{1,2}$ if $r > 0$ and $r \geq -2q\bar{l}_{1,2}$ if $r < 0$ for all $\alpha \in [0, 1]$

If (C11) or (C12) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q, r), (\bar{K}_{2,1}q + 2\bar{k}r\bar{l}_{2,1}, -2q\bar{l}_{1,1} - \bar{K}_{2,1}r)\},$$

where the conditions $(C_i)_{i=7,12}$ are shown in Table 4.

Example 2.3. Consider $A_1 = (1; 2; 3; 4)$ be a symmetric fuzzy number with respect to $x_1 = \frac{5}{2}$, $A_2 = (-3; -2; -1; 0)$ be a symmetric fuzzy number with respect to $x_2 = \frac{-3}{2}$. We can see that $\bar{k} = \frac{x_2}{x_1} = \frac{-3}{5} < 0$.

For $A_1 = (1; 2; 3; 4)$, we have

$$A_{1\alpha}^- = 1 + \alpha, A_{1\alpha}^+ = 4 - \alpha, \text{len}[A_1](\alpha) = 3 - 2\alpha \text{ for all } \alpha \in [0, 1].$$

For $A_2 = (-3; -2; -1; 0)$, we have

$$A_{2\alpha}^- = -3 + \alpha, A_{2\alpha}^+ = -\alpha, \text{len}[A_2](\alpha) = 3 - 2\alpha.$$

We consider $A \in \mathbb{LC}(A_1, A_2)$ with $A = 2A_1 \oplus A_2$. Here, we can see that $q = 2, r = -1$ and

$$\begin{aligned} \bar{K}_1(\alpha) &= \text{len}[A_2](\alpha) - \bar{k}\text{len}[A_1](\alpha) = \frac{8}{5}(3 - 2\alpha) > 0, \\ \bar{K}_2(\alpha) &= \text{len}[A_2](\alpha) + \bar{k}\text{len}[A_2] = \frac{2}{5}(3 - 2\alpha) > 0, \alpha \in [0, 1]. \end{aligned}$$

It implies that $\bar{K}_{2,1} = \frac{1}{4}$ and $l_{2,1} = l_{1,1} = \frac{5}{8}$. From Theorem 2.5, we have

$$\begin{aligned} \psi^{-1}(\psi(q, r)) &= \{(q, r), (\bar{K}_{1,2}q + 2\bar{k}l_{2,1}r, -2q\bar{l}_{1,1} - \bar{K}_{1,2}r)\} \\ &= \left\{ (2; 1), \left(\frac{-1}{4}, \frac{-11}{4} \right) \right\}. \end{aligned}$$

Example 2.4. Consider $A_1 = (0; 1; 2)$ be a symmetric fuzzy number with respect to $x^* = 1$, $A_2 = (-1; 1; 3)$ be a symmetric fuzzy number with respect to $x^* = 1$. We can see that $\bar{k} = 1 > 0$. With $A_1 = (0; 1; 2)$, we have $A_{1\alpha}^- = \alpha, A_{1\alpha}^+ = 2 - \alpha$ and $\text{len}[A_1](\alpha) = 2 - 2\alpha$ for $\alpha \in [0, 1]$. With $A_2 = (-1; 1; 3)$, we have $A_{2\alpha}^- = -1 + 2\alpha, A_{2\alpha}^+ = 3 - 2\alpha$ and $\text{len}[A_2](\alpha) = 4 - 4\alpha$ for all $\alpha \in [0, 1]$. We consider $A \in \mathbb{LC}(A_1, A_2)$ with $A = A_1 \oplus (-1)A_2$. We can see that $q = 1, r = -1, K_3(\alpha) = \text{len}[A_2](\alpha) - \text{len}[A_1](\alpha) = 2 - 2\alpha \geq 0$ for all $\alpha \in [0, 1]$ and

$$r = -1 < \frac{-q}{2kl_{2,2}} = \frac{-1}{4}.$$

Table 5
Conditions for the representation of $\psi^{-1}(\psi(q, r))$ in Theorem 2.6.

Conditions	Description
(D1)	$q > 0, r > 0, 0 < v < \frac{q}{m} + m$ with $m > 0$ or $v \geq \max\{\frac{q}{m} + r; 0\}$ with $m < 0$
(D2)	$q < 0, r < 0, \frac{q}{m} + r < v < 0$ with $m > 0$ or $v < \min\{\frac{q}{m} + r; 0\}$ with $m < 0$
(D3)	$q > 0, r > 0, \frac{-q}{m} - r < v < 0$ with $m > 0$ or $v < \min\{\frac{-q}{m} - r; 0\}$ with $m < 0$
(D4)	$q < 0, r < 0, 0 < v < \frac{-q}{m} - r$ with $m > 0$ or $v > \max\{\frac{-q}{m} - r; 0\}$ with $m < 0$
(D5)	$q > 0, r < 0, 0 < v < \frac{q}{m} - r$ with $m > 0$ or $v > \max\{\frac{q}{m} - r; 0\}$ with $m < 0$
(D6)	$q < 0, r > 0, \frac{q}{m} - r < v < 0$ with $m > 0$ or $v < \min\{\frac{q}{m} - r; 0\}$ with $m < 0$
(D7)	$q > 0, r < 0, \frac{-q}{m} + r < v < 0$ with $m > 0$ or $v < \min\{\frac{-q}{m} + r; 0\}$ with $m < 0$
(D8)	$q < 0, r > 0, 0 < v < \frac{-q}{m} + r$ with $m > 0$ or $v > \max\{\frac{-q}{m} + r; 0\}$ with $m < 0$

From Theorem 2.5, we have $\bar{K}_{1,2}q + 2\bar{kr}\bar{l}_{2,2} = -1$ and $2q\bar{l}_{1,2} - \bar{K}_{1,2}r = 1$, which implies that $\psi^{-1}(1, -1) = \{(1, -1), (-1; 1)\}$.

Theorem 2.4 and Theorem 2.5 have shown the possible cases of $\psi^{-1}(\psi(q, r))$ if $(A_1|x_1), (A_2|x_2)$ with $x_1 \neq 0$. If $(A_1|x_1), (A_2|x_2)$ with $x_1 = x_2 = 0$, the equality $u\rho_{A_1}(\alpha) + v\rho_{A_2}(\alpha) = q\rho_{A_1}(\alpha) + r\rho_{A_2}(\alpha)$ always holds. Thus, the set $\psi^{-1}(\psi(q, r))$ will have infinitely many elements, which is different from the case where the set $\psi^{-1}(\psi(q, r))$ has only two elements.

Theorem 2.6. Let $(A_1|0), (A_2|0)$ and $A = qA_1 \oplus rA_2$ with $(q, r) \in \mathbb{R}^2$. Suppose that $\frac{A_{2\alpha}^+}{A_{1\alpha}^+} = m$ for all $\alpha \in [0, 1]$, then the inverse image of $\psi(q, r)$ is defined as follows:

1) The case $qr \geq 0$.

If (D1) or (D2) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{[q + (r - v)m, v], [-q + (v - r)m, v]\}.$$

If (D3) or (D4) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{[(q + (r + v)m, v)], [(-q - (r + v)m, v)]\}.$$

2) The case $qr < 0$.

If (D5) or (D6) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q - (v + r)m, v), (-q + (v + r)m, v)\}.$$

If (D7) or (D8) is satisfied then

$$\psi^{-1}(\psi(q, r)) = \{(q + (v - r)m, v), (-q + (r - v)m, v)\},$$

where the conditions $(D_i)_{i=1,8}$ are shown in Table 5.

Proof. Assume that there exist $u, v \in \mathbb{R}$ such that $qA_1 \oplus rA_2 = uA_1 \oplus vA_2$.

To prove this theorem, we consider two cases: $qr > 0$ and $qr < 0$. Since the steps of the argument are similar, we will prove in detail the two cases when $q > 0, r > 0, u > 0, v > 0$ or $q < 0, r < 0, u < 0, v < 0$ and $q > 0, r > 0, u < 0, v > 0$ or $q < 0, r < 0, u > 0, v < 0$. The remaining cases are deduced similarly.

Case 1: $q > 0, r > 0, u > 0, v > 0$ and $q < 0, r < 0, u < 0, v < 0$. We can see that u and v satisfy the system

$$\begin{cases} u\text{len}[A_1](\alpha) + v\text{len}[A_2](\alpha) = q\text{len}[A_1](\alpha) + r\text{len}[A_2](\alpha), \text{ for all } \alpha \in [0, 1]. \\ u\rho_{A_1}(\alpha) + v\rho_{A_2}(\alpha) = q\rho_{A_1}(\alpha) + r\rho_{A_1}(\alpha). \end{cases}$$

From $(A_1|0)$ and $(A_2|0)$, we get $\rho_{A_1}(\alpha) = \rho_{A_2}(\alpha) = 0$. For this reason, we get

$$u\text{len}[A_1](\alpha) + v\text{len}[A_2](\alpha) = q\text{len}[A_1](\alpha) + r\text{len}[A_2](\alpha) \Leftrightarrow uA_{1\alpha}^+ + vA_{2\alpha}^+ = qA_{1\alpha}^+ + rA_{2\alpha}^+.$$

Because there exists $m \in \mathbb{R} \setminus \{0\}$ such that $\frac{A_{2\alpha}^+}{A_{1\alpha}^+} = m$, we have $u = q + (r - v)m, v \in \mathbb{R}$.

In the case when $q > 0, r > 0, u > 0, v > 0$, in order to get $u > 0, v > 0$, we need $0 < v < \frac{q}{m} + r$ with $m > 0$ and $v > \max\{\frac{q}{m} + r; 0\}$ with $m < 0$. In the case when $q < 0, r < 0, u < 0, v < 0$, in order to get $u < 0, v < 0$, we need $\frac{q}{m} + r < v < 0$ with $m > 0$ and $v < \min\{\frac{q}{m} + r; 0\}$ with $m < 0$.

Case 2: $q > 0, r > 0, u < 0, v > 0$ or $q < 0, r < 0, u > 0, v < 0$. The argument is similar to Case 1, we have

$$-u\text{len}[A_1](\alpha) + v\text{len}[A_2](\alpha) = q\text{len}[A_1](\alpha) + r\text{len}[A_2](\alpha). \Leftrightarrow -uA_{1\alpha}^+ + vA_{2\alpha}^+ = qA_{1\alpha}^+ + rA_{2\alpha}^+.$$

This leads to $u = -q + (v - r)m, v \in \mathbb{R}$. In the case when $q > 0, r > 0, u < 0, v > 0$, in order to get $u < 0, v > 0$, we need $0 < v < \frac{q}{m} + r$ with $m > 0$ and $v > \max\{\frac{q}{m} + r; 0\}$ with $m < 0$. In the case when $q < 0, r < 0, u > 0, v < 0$, in order to get $u > 0, v < 0$, we need $\frac{q}{m} + r < v < 0$ with $m > 0$ and $v > \max\{\frac{q}{m} + r; 0\}$ with $m < 0$.

Therefore, when condition (D1) or (D2) is satisfied, by combining the results from both cases 1 and 2, we obtain $\psi^{-1}(\psi(q, r)) = \{(q + (r - v)m, v), (-q + (v - r)m, v)\}$.

By applying similar arguments to the remaining cases of $qr > 0$ and the case $qr < 0$, we get the desired result. $\square$

Example 2.5. Consider $A_1 = (-1; 0; 1)$ be a symmetric fuzzy number with respect to $x^* = 0$, $A_1 = (-3; 0; 3)$ be a symmetric fuzzy number with respect to $x^* = 0$. With $A_1 = (-1; 0; 1)$, we have $A_{1\alpha}^- = -1 + \alpha, A_{1\alpha}^+ = 1 - \alpha$ and $\text{len}[A_1](\alpha) = 2 - 2\alpha$ for $\alpha \in [0, 1]$.

With $A_2 = (-3; 0; 3)$, we have $A_{2\alpha}^- = -3 + 3\alpha, A_{2\alpha}^+ = 3 - 3\alpha$ for all $\alpha \in [0, 1]$. Therefore, $\frac{A_{2\alpha}^+}{A_{1\alpha}^+} = 3 > 0$. We consider $A \in \mathbb{LC}(A_1, A_2)$

with $A = (-3)A_1 \oplus A_2 = (-6; 0; 6)$. We can see that $q = -3, r = 1$ and $\frac{q}{m} - r = -2$. From Theorem 2.6, we have $\psi^{-1}(\psi(-3; 1)) = \{(3v + 6; v), (-3v - 6; v)\}$ with $-2 < v < 0$. Indeed, if we choose $v = -1$ or $v = \frac{-1}{3}$, we can see that $\psi(u, v) = (-6; 0; 6)$.

3. Calculus for generalized linearly correlated fuzzy processes

3.1. The Fréchet derivative of linearly correlated fuzzy processes

Let $J = [a, b]$ be a closed interval and $A_1, A_2 \in \mathbb{R}_F$. The function

$$g : [a, b] \rightarrow \mathbb{LC}(A_1, A_2)$$

is called a linearly correlated fuzzy number-valued function. In fact, assume that $g(t)$ is a linearly correlated fuzzy number-valued function on $[a, b]$, then there exists a pair of real-valued functions $q, r : [a, b] \rightarrow \mathbb{R}$ such that $g(t) = q(t)A_1 \oplus r(t)A_2$ for any $t \in [a, b]$. Specifically, if $\{A_1, A_2\}$ is SLI, then the pair of functions $q(t)$ and $r(t)$ is unique. However, if $\{A_1, A_2\}$ is not SLI, then it is not unique. We denote by $\mathcal{L}(J, \mathbb{LC}(A_1, A_2))$ the set of all continuous linear operators from J to $\mathbb{LC}(A_1, A_2)$, where $J = [a, b]$ with $a, b \in \mathbb{R}$.

Definition 3.1 ([16]). The function $g : J \rightarrow \mathbb{LC}(A_1, A_2)$ is Fréchet differentiable at $t \in J$ if there is a linear operator $T \in \mathcal{L}(J, \mathbb{LC}(A_1, A_2))$ such that

$$\lim_{h \rightarrow 0} \frac{\|g(t+h) - g(t) - T(h)\|_\psi}{|h|} = 0.$$

Then, the linear operator T is called the Fréchet derivative of g at t and denoted by $T = g'_T(t)$.

The mapping g is Fréchet differentiable on J if and only if it is Fréchet differentiable at every point $t \in J$.

3.1.1. The case when $\{A_1, A_2\}$ is SLI

When $\{A_1, A_2\}$ is SLI, ψ is an isomorphism with the norm $\|B\|_\psi = \|\psi^{-1}(B)\|$ for any $B = \psi(q_B, r_B) \in \mathbb{LC}(A_1, A_2)$. Thus, we can conclude that both ψ and ψ^{-1} are continuous linear operators and, according to [17] both operators ψ and ψ^{-1} are Fréchet differentiable.

Lemma 3.1. Let $\{A_1, A_2\}$ be SLI and $g : J \rightarrow \mathbb{LC}(A_1, A_2)$. The function g is Fréchet differentiable at $t \in \mathbb{R}$ if and only if $\psi^{-1} \circ g$ is Fréchet differentiable at t .

Proof. Assume that g is Fréchet differentiable at t and ψ_A^{-1} is Fréchet differentiable at $g(t)$. By applying the chain rule in [17], we can establish that the composite mapping $\psi^{-1} \circ g$ is also Fréchet differentiable at t . Conversely, let $p = \psi^{-1} \circ g$ and suppose that p is Fréchet differentiable at $t \in \mathbb{R}$. Since $\psi \circ p = g$ and ψ is Fréchet differentiable, we have that g is Fréchet differentiable at t . $\square$

The following lemma establishes that a function $g : \mathbb{R} \rightarrow \mathbb{LC}(A_1, A_2)$ can be written as the composition of ψ with a single function $p : \mathbb{R} \rightarrow \mathbb{R}^2$, given by $p = \psi^{-1} \circ g$.

Lemma 3.2. Let $\{A_1, A_2\}$ be SLI. For each function $g : \mathbb{R} \rightarrow \mathbb{LC}(A_1, A_2)$, there exist unique functions $q, r : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(t) = \psi(q(t), r(t))$ for all $t \in \mathbb{R}$. Additionally, $(q, r) = \psi^{-1} \circ g$.

Proof. Consider the function $h = \psi^{-1} \circ g$, which maps from $\mathbb{R}$ to $\mathbb{R}^2$. Given that ψ is bijective, we can establish the relationship $g(t) = (\psi \circ h)(t)$ for all $t \in \mathbb{R}$. Assume that there exists a function $p : \mathbb{R} \rightarrow \mathbb{R}^2$ such that $g = \psi \circ p$ for all $t \in \mathbb{R}$. Then, $g(t) = \psi(p(t))$ implies $h(t) = \psi^{-1}(g(t)) = p(t)$. Furthermore, since $h : \mathbb{R} \rightarrow \mathbb{R}^2$ can be represented as a vector of two real functions q and r , such that $h(t) = (q(t), r(t))$ for all $t \in \mathbb{R}$, it follows that $g(t) = (\psi \circ h)(t) = \psi((q(t), r(t)))$ for all $t \in \mathbb{R}$. $\square$

The following theorem establishes a significant methodological reduction: the problem of calculating the Fréchet derivative of $g : \mathbb{R} \rightarrow \mathbb{LC}(A_1, A_2)$ for any $t \in \mathbb{R}$ can be systematically reduced to calculating the derivative of a function from $\mathbb{R}$ to $\mathbb{R}^2$.

Theorem 3.1. Let $\{A_1, A_2\}$ be SLI, $q, r : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{LC}(A_1, A_2)$ such that $g(t) = \psi(q(t), r(t))$ for all $t \in \mathbb{R}$. The function g is Fréchet differentiable at $t \in \mathbb{R}$ if and only if $q'(t)$ and $r'(t)$ exist for all $t \in \mathbb{R}$. Additionally, $g'_A(t) = \psi(q'(t), r'(t))$.

Proof. Let $p(t) = (q(t), r(t))$ for all $t \in \mathbb{R}$. Assuming the existence of derivatives $q'(t)$ and $r'(t)$ for $t \in \mathbb{R}$, the Fréchet derivative of p at t emerges as $p'_A(t) = (q'(t), r'(t))$ for all $t \in \mathbb{R}$. Applying the chain rule in [17], we can derive the Fréchet derivative of the composite mapping as $g'_A(t) = (\psi \circ p'_A)(t) = \psi(q'(t), r'(t))$ for all $t \in \mathbb{R}$. Conversely, suppose g is Fréchet differentiable at $t \in \mathbb{R}$. According to Lemma 3.2, $p = \psi^{-1} \circ g$ is Fréchet differentiable at t . This is equivalent to $q(t)$ and $r(t)$ also being Fréchet differentiable. Then, there exist linear operators $T_q, T_r \in \mathcal{L}(\mathbb{R}, \mathbb{R})$ such that

$$\lim_{h \rightarrow 0} \frac{\|q(t+h) - q(t) - T_q(h)\|}{|h|} = 0$$

$$\lim_{h \rightarrow 0} \frac{\|r(t+h) - r(t) - T_r(h)\|}{|h|} = 0$$

This means

$$q(t+h) - q(t) = T_q(h) + o(h)$$

$$r(t+h) - r(t) = T_r(h) + o(h)$$

Since $\mathbb{R}$ is a Hilbert space, the Riesz representation theorem guarantees that the linear functions $T_q, T_r : \mathbb{R} \rightarrow \mathbb{R}$ can be expressed as $T_q(h) = a_q h$ and $T_r(h) = a_r h$ for some $a_q, a_r \in \mathbb{R}$. Applying these insights yields:

$$q'(t) = \lim_{h \rightarrow 0} \frac{q(t+h) - q(t)}{h} = a_q$$

$$r'(t) = \lim_{h \rightarrow 0} \frac{r(t+h) - r(t)}{h} = a_r$$

So, if g is Fréchet differentiable at $t \in \mathbb{R}$, then $q'(t)$ and $r'(t)$ exist. $\square$

3.1.2. The case when $\{A_1, A_2\}$ is not SLI

Let us now consider the case when $\{A_1, A_2\}$ is not SLI.

Theorem 3.2. Let A_1, A_2 be non-symmetric fuzzy numbers and suppose there exists $k \in \mathbb{R} \setminus \{0\}$ such that $\rho_{A_2}(\alpha) = k \rho_{A_1}(\alpha)$ for all $\alpha \in [0, 1]$. Suppose that there exist continuous functions $q_i, r_i : D \rightarrow \mathbb{R}$, where for $i = \overline{1, 2}$ and D is an open subset of $\mathbb{R}$, satisfying $q_1(t)r_1(t) \neq 0$, and $\psi(q_1(t), r_1(t)) = \psi(q_2(t), r_2(t))$ for all $t \in D$. Then, if $q_1(t)$ and $r_1(t)$ have the same sign for all $t \in D$, the following assertions hold If (E1) or (E2) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = q_1(t)K_{1,2}, r_2(t) = 2q_1(t)l_{1,2} + r_1(t).$$

If (E3) or (E4) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = q_1(t) + 2kr_1(t)l_{2,2}, r_2(t) = -K_{1,2}r_1(t).$$

If (E5) or (E6) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = K_{2,1}q_1(t) + 2kl_{2,1}r_1(t), r_2(t) = -2l_{1,1}q_1(t) - K_{2,1}r_1(t),$$

where the conditions $(E_i)_{i=\overline{1,6}}$ are shown in Table 6.

Proof. For $t_0 \in D$, since q_1, r_1, q_2, r_2 are continuous, there exist $\delta_1 > 0, \delta_2 > 0, \delta_3 > 0, \delta_4 > 0$ such that $q_1(t)$ has the same sign of $q_1(t_0)$ for all $t \in B(t_0, \delta_1) := \{t \in \mathbb{R} | |t - t_0| < \delta_1\} \subseteq D$, $r_1(t)$ has the same sign of $r_1(t_0)$ for all $t \in B(t_0, \delta_2) := \{t \in \mathbb{R} | |t - t_0| < \delta_2\} \subseteq D$, $q_2(t)$ has the same sign of $q_2(t_0)$ for all $t \in B(t_0, \delta_3) := \{t \in \mathbb{R} | |t - t_0| < \delta_3\} \subseteq D$ and $r_2(t)$ has the same sign of $r_1(t_0)$ for all $t \in B(t_0, \delta_4) :=$

Table 6

Conditions for the expressions of $q_i(t)$, $r_i(t)$ ($i = 1, 2$) when non-symmetric fuzzy numbers A_1, A_2 are not SLI in Theorem 3.2.

Conditions	Description
(E1)	$k > 0, K_1(\alpha) \leq 0$
(E2)	$k < 0, K_2(\alpha) < 0, r_1(t) > -2q_1(t)l_{1,2}$ if $r_1(t) \geq 0$ and $r_1(t) < -2q_1(t)l_{1,2}$ if $r_1(t) < 0$
(E3)	$k > 0, K_1(\alpha) \geq 0$
(E4)	$k < 0, K_2(\alpha) > 0, r_1(t) \leq \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) > 0$ and $r_1(t) > \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) < 0$
(E5)	$k < 0, K_2(\alpha) > 0, r_1(t) > \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) > 0$ and $r_1(t) \leq \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) < 0$
(E6)	$k < 0, K_2(\alpha) < 0, r_1(t) < -2q_1(t)l_{1,2}$ if $r_1(t) > 0$ and $r_1(t) \geq -2q_1(t)l_{1,2}$ if $r_1(t) < 0$

$\{t \in \mathbb{R} \mid |t - t_0| < \delta_4\} \subseteq D$. Set $\delta = \min\{\delta_1, \delta_2, \delta_3, \delta_4\}$ and $t \in B(t_0, \delta)$, we can see that the values $q_i(t), r_i(t), i = 1, 2$ keep the same sign. Similar to Theorem 2.1, we have the following cases:

Case 1: $q_1(t) > 0, r_1(t) > 0, q_2(t) > 0, r_2(t) > 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) < 0, r_2(t) < 0$ for all $t \in B(t_0, \delta)$.

Since $q_1(t)A_1 \oplus r_1(t)A_2 = q_2(t)A_1 \oplus r_2(t)A_2$, we have $q_1(t)\text{len}[A_1](\alpha) + r_1(t)\text{len}[A_2](\alpha) = q_2(t)\text{len}[A_1](\alpha) + r_2(t)\text{len}[A_2](\alpha)$, for all $t \in B(t_0, \delta)$. Therefore, $q_2(t)$ and $r_2(t)$ satisfy the following system

$$\begin{cases} q_2(t)\text{len}[A_1](\alpha) + r_2(t)\text{len}[A_2](\alpha) = q_1(t)\text{len}[A_1](\alpha) + r_1(t)\text{len}[A_2](\alpha), \\ q_2(t) + kr_2(t) = q_1(t) + kr_1(t). \end{cases}$$

We can obtain $q_2(t) = q_1(t), r_2(t) = r_1(t)$.

Case 2: $q_1(t) > 0, r_1(t) > 0, q_2(t) < 0, r_2(t) > 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) > 0, r_2(t) < 0$ for all $t \in B(t_0, \delta)$.

Since $q_1(t)A_1 \oplus r_1(t)A_2 = q_2(t)A_1 \oplus r_2(t)A_2$, we get that $-q_2(t)\text{len}[A_1](\alpha) + r_2(t)\text{len}[A_2](\alpha) = q_1(t)\text{len}[A_1](\alpha) + r_1(t)\text{len}[A_2](\alpha)$ for all $t \in B(t_0, \delta)$. So, $q_2(t)$ and $r_2(t)$ satisfy the following system

$$\begin{cases} -q_2(t)\text{len}[A_1](\alpha) + r_2(t)\text{len}[A_2](\alpha) = q_1(t)\text{len}[A_1](\alpha) + r_1(t)\text{len}[A_2](\alpha), \\ q_2(t) + kr_2(t) = q_1(t) + kr_1(t). \end{cases}$$

We can obtain $q_2(t) = K_{1,2}q_1(t), r_2(t) = 2l_{1,2}q_1(t) + r_1(t), t \in B(t_0, \delta)$.

The argument is similar to case 2 in Theorem 2.1, we get that if $q(t)r(t) > 0$ and the condition (E1) or (E2) satisfies then

$$\begin{cases} q_2(t) = q_1(t) \\ r_2(t) = r_1(t) \end{cases} \quad \text{or} \quad \begin{cases} q_2(t) = K_{1,2}q_1(t), \\ r_2(t) = 2l_{1,2}q_1(t) + r_1(t), t \in B(t_0, \delta) \end{cases}$$

If the condition (E3) or (E4), (E5) or (E6) are satisfied, by the same argument we get the result as given in the theorem. $\square$

Remark 3.1. Following a similar proof to that of Theorem 3.2, by considering different cases of $q_i(t), r_i(t)$ and the symmetry of A_1, A_2 we obtain the following results:

a. Let A_1, A_2 be non-symmetric fuzzy numbers and suppose there exists $k \in \mathbb{R} \setminus \{0\}$ such that $\rho_{A_2}(\alpha) = k\rho_{A_1}(\alpha)$ for all $\alpha \in [0, 1]$. Suppose that there exist continuous functions $q_i, r_i: D \rightarrow \mathbb{R}$, where for $i = \overline{1, 2}$ and D is an open subset of $\mathbb{R}$, satisfying $q_1(t)r_1(t) \neq 0$ and $\psi(q_1(t), r_1(t)) = \psi(q_2(t), r_2(t))$ for all $t \in D$. Then, if $q_1(t)$ and $r_1(t)$ have the opposite sign for all $t \in D$, the following assertions hold

If (E7) or (E8) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = K_{2,1}q_1(t), r_2(t) = -2l_{1,1}q_1(t) + r_1(t).$$

If (E9) or (E10) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = q_1(t) + 2kl_{2,1}r_1(t), r_2(t) = -K_{2,1}r_1(t).$$

If (E11) or (E12) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = K_{1,2}q_1(t) + 2kl_{2,2}r_1(t), r_2(t) = 2l_{1,2}q_1(t) - K_{1,2}r_1(t),$$

where the conditions $(E_i)_{i=7,12}$ are shown in Table 7.

b. Let $(A_1|x_1), (A_2|x_2), x_1 \neq 0$ and an open set $D \subseteq \mathbb{R}$. Suppose that there exist continuous functions $q_i, r_i: D \rightarrow \mathbb{R}$ for all $i = \overline{1, 2}$ such that $q_1(t)r_1(t) \neq 0$ and $\psi(q_1(t), r_1(t)) = \psi(q_2(t), r_2(t))$ for all $t \in D$. Then, if $q_1(t)$ and $r_1(t)$ have the opposite sign for all $t \in D$, the following assertions hold

If (H1) or (H2) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = q_1(t)\bar{K}_{1,2}, r_2(t) = 2q_1(t)\bar{l}_{1,2} + r_1(t).$$

Table 7

Conditions for the expressions of $q_i(t)$, $r_i(t)$ ($i = 1, 2$) when non-symmetric fuzzy numbers A_1, A_2 are not SLI in the case (a) of Remark 3.1.

Conditions	Description
(E7)	$k < 0, K_2(\alpha) \leq 0$
(E8)	$k > 0, K_1(\alpha) < 0, r_1(t) \geq 2l_{1,1}q_1(t)$ if $r_1(t) > 0$ and $r_1(t) < 2l_{1,1}q_1(t)$ if $r_1(t) < 0$
(E9)	$k < 0, K_2(\alpha) > 0$
(E10)	$k > 0, K_1(\alpha) > 0, r_1(t) < \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) > 0$ and $r_1(t) \geq \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) < 0$
(E11)	$k > 0, K_1(\alpha) < 0, r_1(t) < 2l_{1,1}q_1(t)$ if $r_1(t) > 0$ and $r_1(t) \geq 2l_{1,2}q_1(t)$ if $r_1(t) < 0$
(E12)	$k > 0, K_1(\alpha) > 0, r_1(t) \geq \frac{-q_1(t)}{2kl_{2,1}}$ if $r_1(t) > 0$ and $r_1(t) < \frac{-q_1(t)}{2kl_{2,1}}$ if $r_1(t) < 0$

Table 8

Conditions for the expressions of $q_i(t)$, $r_i(t)$ ($i = 1, 2$) when symmetric fuzzy numbers A_1, A_2 are not SLI in the case (b) of Remark 3.1.

Conditions	Description
(H1)	$\bar{k} > 0, \bar{K}_1(\alpha) \leq 0,$
(H2)	$\bar{k} < 0, \bar{K}_2(\alpha) < 0, r_1(t) \geq -2q_1(t)l_{1,2}$ if $r_1(t) > 0$ and $r_1(t) < -2q_1(t)l_{1,2}$ if $r_1(t) < 0$
(H3)	$\bar{k} > 0, \bar{K}_1(\alpha) > 0$
(H4)	$\bar{k} < 0, \bar{K}_2(\alpha) < 0, r_1(t) < \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) > 0$ and $r_1(t) > \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) < 0$
(H5)	$\bar{k} < 0, \bar{K}_2(\alpha) > 0, r_1(t) > \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) > 0$ and $r_1(t) < \frac{-q_1(t)}{2kl_{2,2}}$ if $r_1(t) < 0$
(H6)	$\bar{k} < 0, \bar{K}_2(\alpha) < 0, r_1(t) < -2q_1(t)l_{1,2}$ if $r_1(t) > 0$ and $-2q_1(t)l_{1,2} < r_1(t)$ if $r_1(t) < 0$

If (H3) or (H4) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = q_1(t) + 2\bar{k}r_1(t)\bar{l}_{2,2}, r_2(t) = -\bar{K}_{1,2}r_1(t).$$

If (H5) or (H6) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = \bar{K}_{2,1}q_1(t) + 2\bar{k}l_{2,1}r_1(t), r_2(t) = -2\bar{l}_{1,1}q_1(t) - \bar{K}_{2,1}r_1(t),$$

where the conditions $(H_i)_{i=1,6}$ are shown in Table 8.

c. Let $(A_1|x_1), (A_2|x_2), x_1 \neq 0$ and an open set $D \subseteq \mathbb{R}$. Suppose that there exist continuous functions $q_i, r_i: D \rightarrow \mathbb{R}$, for all $i = 1, 2$ such that $q_1(t)r_1(t) \neq 0$, and $\psi(q_1(t), r_1(t)) = \psi(q_2(t), r_2(t))$ for all $t \in D$. Then, if $q_1(t)$ and $r_1(t)$ have the same sign for all $t \in D$, the following assertions hold

If (H7) or (H8) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = \bar{K}_{2,1}q_1(t), r_2(t) = -2\bar{l}_{1,1}q_1(t) + r_1(t).$$

If (H9) or (H10) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = q_1(t) + 2\bar{k}l_{2,1}r_1(t), r_2(t) = -\bar{K}_{2,1}r_1(t).$$

If (H11) or (H12) is satisfied then

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = \bar{K}_{1,2}q_1(t) + 2\bar{k}l_{2,2}r_1(t), r_2(t) = 2\bar{l}_{1,2}q_1(t) - \bar{K}_{1,2}r_1(t),$$

where the conditions $(H_i)_{i=6,12}$ are shown in Table 9.

The next theorem motivates the extension of the notion of Fréchet derivative for fuzzy functions of the form $f(t) = \psi(q(t), r(t))$ for all $t \in \mathbb{R}$ when $\{A_1, A_2\}$ is not SLI and therefore, $\mathbb{L}\mathbb{C}(A_1, A_2)$ is not a Banach space. These results allow us to define the Fréchet derivative of f in terms of the derivatives of q and r such as in Theorem 3.4 that provides a practical formula for the previous case where $\{A_1, A_2\}$ is not SLI. In addition, this definition does not depend on the choice of the functions q and r .

Theorem 3.3. Let A_1, A_2 be fuzzy numbers, distinct from real numbers, such that $\{A_1, A_2\}$ is not SLI. If there exist two functions $q_i, r_i: \mathbb{R} \rightarrow \mathbb{R}$ that are differentiable at $t_0 \in \mathbb{R}$ for $i = 1, 2$ such that $q_1(t)r_1(t) \neq 0$ and $\psi(q_1(t), r_1(t)) = \psi(q_2(t), r_2(t))$ for all $t \in B(t_0, \delta), \delta > 0$ then

$$\psi(q'_1(t_0), r'_1(t_0)) = \psi(q'_2(t_0), r'_2(t_0)).$$

Proof. Without loss of generality, we will prove this theorem in the case where A_1, A_2 are non-symmetric fuzzy numbers and there exists $k \in \mathbb{R} \setminus \{0\}$ such that $\rho_{A_2}(\alpha) = k\rho_{A_1}(\alpha)$ for all $\alpha \in [0, 1]$ ($\{A_1, A_2\}$ is not SLI). From Theorem 3.2, we have 3 cases corresponding

Table 9

Conditions for the expressions of $q_i(t)$, $r_i(t)$ ($i = 1, 2$) when symmetric fuzzy numbers A_1, A_2 are not SLI in the case (c) of Remark 3.1.

Conditions	Description
(H7)	$\bar{k} < 0, \bar{K}_2(\alpha) \leq 0$
(H8)	$\bar{k} > 0, \bar{K}_1(\alpha) < 0, r_1(t) \geq 2\bar{l}_{1,1}q_1(t)$ if $r_1(t) > 0$ and $r_1(t) < 2\bar{l}_{1,1}q_1(t)$ if $r_1(t) < 0$
(H9)	$\bar{k} < 0, \bar{K}_2(\alpha) > 0$
(H10)	$\bar{k} > 0, \bar{K}_1(\alpha) > 0, r_1(t) < \frac{-q_1(t)}{2\bar{k}l_{2,2}}$ if $r_1(t) > 0$ and $r_1(t) \geq \frac{-q_1(t)}{2\bar{k}l_{2,2}}$ if $r_1(t) < 0$
(H11)	$\bar{k} > 0, \bar{K}_1(\alpha) < 0, r_1(t) < 2\bar{l}_{1,1}q_1(t)$ if $r_1(t) > 0$ and $r_1(t) \geq 2\bar{l}_{1,2}q_1(t)$ if $r_1(t) < 0$
(H12)	$k > 0, \bar{K}_1(\alpha) > 0, r_1(t) \geq \frac{-q_1(t)}{2\bar{k}l_{2,1}}$ if $r_1(t) > 0$ and $r_1(t) < \frac{-q_1(t)}{2\bar{k}l_{2,1}}$ if $r_1(t) < 0$

to some conditions $(E_i), i = 1, \dots, 6$. Without loss of generality, we will prove this theorem for the first case. The remaining cases can be proved similarly.

From Theorem 3.2, if the condition (E1) or (E2) is satisfied, for each $t \in B(t_0, \delta)$, we have

$$q_2(t) = q_1(t), r_2(t) = r_1(t) \text{ or } q_2(t) = q_1(t)K_{1,2}, r_2(t) = 2q_1(t)l_{1,2} + r_1(t).$$

Thus, there exists a sequence (t_n) in $B(t_0, \delta) \setminus \{t_0\}$ that converges to t_0 and satisfies one of the following conditions:

- (i) $q_2(t_n) = q_1(t_n)$ and $r_2(t_n) = r_1(t_n)$,
- (ii) $q_2(t_n) = q_1(t_n)K_{1,2}$ and $r_2(t_n) = 2q_1(t_n)l_{1,2} + r_1(t_n)$

for all $n \in \mathbb{N}$. Firstly, we will point out that for all $n \in \mathbb{N}$, the sequence (t_n) satisfies either condition (i) or condition (ii), as separate cases. Indeed, suppose that there exists a sequence $\{t_n\}$ for which conditions (i) and (ii) are satisfied alternately. Then, let $\{t_{n_m}\}$ and $\{t_{n_k}\}$ be subsequences of $\{t_n\}$ such that $\{t_{n_m}\}$ satisfies (i), meaning $q_2(t_{n_m}) = q_1(t_{n_m})$ for all $m \in \mathbb{N}$, and $\{t_{n_k}\}$ satisfies (ii), meaning $q_2(t_{n_k}) = q_1(t_{n_k})K_{1,2}$ for all $k \in \mathbb{N}$. Since q_1 and q_2 are continuous functions, and the sequences $(t_{n_m}), (t_{n_k})$ converge to t_0 , we have

- $\lim_{m \rightarrow \infty} q_1(t_{n_m}) = q_1(t_0)$ and $\lim_{m \rightarrow \infty} q_2(t_{n_m}) = q_2(t_0)$.
- $\lim_{k \rightarrow \infty} q_1(t_{n_k}) = q_1(t_0)$ and $\lim_{k \rightarrow \infty} q_2(t_{n_k}) = q_2(t_0)$.

This leads to $q_2(t_0) = q_1(t_0)$ and $q_2(t_0) = q_1(t_0)K_{1,2}$. From this, we get $q_1(t_0) = 0$. This contradicts our initial assumption that $q_1(t_0) \neq 0$. Therefore, for all n , we can assert the existence of a sub-sequence t_n converging to t_0 that satisfies one of the two conditions (i) or (ii) for all $n \in \mathbb{N}$.

If the case (i) holds then $q_1(t_0) = q_2(t_0)$ and $r_1(t_0) = r_2(t_0)$ (since q_1, q_2, r_1, r_2 are continuous at t_0). Moreover, we have that

$$q'_1(t_0) = \lim_{n \rightarrow \infty} \frac{q_1(t_n) - q_1(t_0)}{t_n - t_0} = \lim_{n \rightarrow \infty} \frac{q_2(t_n) - q_2(t_0)}{t_n - t_0} = q'_2(t_0)$$

and

$$r'_1(t_0) = \lim_{n \rightarrow \infty} \frac{r_1(t_n) - r_1(t_0)}{t_n - t_0} = \lim_{n \rightarrow \infty} \frac{r_2(t_n) - r_2(t_0)}{t_n - t_0} = r'_2(t_0).$$

Therefore, $\psi(q'_1(t_0), r'_1(t_0)) = \psi(q'_2(t_0), r'_2(t_0))$. On the other hand, if the sequence (t_n) satisfies the condition (ii), then the continuity of q_i, r_i at $t_0, i = 1, 2$ implies that $q_2(t_0) = q_1(t_0)K_{1,2}$ and $r_2(t_0) = 2q_1(t_0)l_{1,2} + r_1(t_0)$. The last observation and the hypothesis that the functions q_i, r_i are differentiable at t_0 for $i = 1, 2$ produce the following equalities:

$$q'_1(t_0) = \lim_{n \rightarrow \infty} \frac{K_{1,2}(q_1(t_n) - q_1(t_0))}{t_n - t_0} = K_{1,2} \lim_{n \rightarrow \infty} \frac{q_2(t_n) - q_2(t_0)}{t_n - t_0} = K_{1,2}q'_2(t_0)$$

and

$$\begin{aligned} r'_1(t_0) &= \lim_{n \rightarrow \infty} \frac{r_1(t_n) - r_1(t_0)}{t_n - t_0} \\ &= 2l_{1,2} \lim_{n \rightarrow \infty} \frac{q_1(t_n) - q_1(t_0)}{t_n - t_0} + \lim_{n \rightarrow \infty} \frac{r_2(t_n) - r_2(t_0)}{t_n - t_0} \\ &= 2l_{1,2}q'_2(t_0) + r'_2(t_0). \end{aligned}$$

Finally, an application of Theorem 3.2 reveals that

$$\psi(q'_1(t_0), r'_1(t_0)) = \psi(K_{1,2}q'_2(t_0), 2l_{1,2}q'_2(t_0) + r'_2(t_0)) = \psi(q'_2(t_0), r'_2(t_0)),$$

which completes the proof. The case where $(A_1|x_1), (A_2|x_2), x_1 \neq 0$ can be proved similarly by applying Remark 3.1. $\square$

Theorem 3.4. Let $\{A_1, A_2\}$ be not SLI and $g(t) = \psi(q(t), r(t))$. If q and r are differentiable in $\mathbb{R}$, then the function g is Fréchet differentiable for every $t \in \mathbb{R}$ and $g'_F(t) = \psi(q'(t), r'(t))$.

Proof. From Theorem 2.3 to Theorem 2.6, we have $\psi^{-1}(\psi(q, r)) = \{(q, r), (u(q, r), v(q, r))\}$ where $u(q, r)$ and $v(q, r)$ are functions of q and r . However, since the functions $u(q, r)$ and $v(q, r)$ are differentiable due to the fact that $q(t)$ and $r(t)$ are differentiable in $\mathbb{R}$, we will prove the theorem in the case where the inverse mapping of ψ is (q, r) . Let $g(t) = \psi(q(t), r(t))$ be a function from $\mathbb{R}$ to $\mathbb{LC}(A_1, A_2)$ such that $q(t)$ and $r(t)$ are differentiable at $t \in \mathbb{R}$ and suppose that the space $\mathbb{R}^2$ is equipped with the maximum norm, i.e., $\|(x, y)\|_\infty = \max\{|x|, |y|\}$.

Given $\epsilon > 0$, let $\gamma(h) = \psi(q(t+h), r(t+h)) - \psi(q(t), r(t)) - \psi(q'(t)h, r'(t)h)$. Since $q(t)$ and $r(t)$ are differentiable, there exists a sufficiently small $\delta > 0$, such that for all $|h| < \delta$, we have

$$\begin{aligned} \left\| \frac{\gamma(h)}{h} \right\|_\psi &= \left\| \frac{\psi(q(t+h), r(t+h)) \ominus \psi(q(t), r(t)) \ominus \psi(q'(t)h, r'(t)h)}{h} \right\|_\psi \\ &= \left\| \left(\frac{q(t+h) - q(t)}{h} - q'(t), \frac{r(t+h) - r(t)}{h} - r'(t) \right) \right\|_\infty \\ &= \max \left\{ \left| \frac{q(t+h) - q(t)}{h} - q'(t) \right|, \left| \frac{r(t+h) - r(t)}{h} - r'(t) \right| \right\} < \epsilon. \end{aligned}$$

So, the function g is Fréchet differentiable at $t \in \mathbb{R}$ and $g'_F(t) = \psi(q'(t), r'(t)), t \in \mathbb{R}$. $\square$

Example 3.1. Consider $A_1 = \left(-1; \frac{1}{2}; \frac{5}{2}\right)$ and $A_2 = (0; 1; 3)$ to be non-symmetric fuzzy numbers. For $A_1 = \left(-1; \frac{1}{2}; \frac{5}{2}\right)$, we have

$$A_{1\alpha}^- = -1 + \frac{3}{2}\alpha, A_{1\alpha}^+ = \frac{5}{2} - 2\alpha, \text{len}[A_1](\alpha) = \frac{7}{2} - \frac{7}{2}\alpha, \rho_{A_1}(\alpha) = \frac{3}{2} - \frac{1}{2}\alpha \text{ for all } \alpha \in [0, 1].$$

For $A_2 = (0; 1; 3)$, we have

$$A_{2\alpha}^- = \alpha, A_{2\alpha}^+ = 3 - 2\alpha, \text{len}[A_2](\alpha) = 3 - 3\alpha, \rho_{A_2}(\alpha) = 3 - \alpha \text{ for all } \alpha \in [0, 1].$$

We can see that $\rho_{A_2}(\alpha) = 2\rho_{A_1}(\alpha)$ for all $\alpha \in [0, 1]$. From Theorem 3.2, this implies

$$k = 2 > 0, K_1(\alpha) = \text{len}[A_2](\alpha) - 2\text{len}[A_1](\alpha) = -4 + 4\alpha \leq 0 \text{ for all } \alpha \in [0, 1].$$

$$K_2(\alpha) = \text{len}[A_2](\alpha) + 2\text{len}[A_1](\alpha) = 10 - 10\alpha \leq 0 \text{ for all } \alpha \in [0, 1].$$

It follows that $l_{1,2} = \frac{\text{len}[A_1](\alpha)}{K_2(\alpha)} = \frac{7}{20}$ and $K_{1,2} = \frac{K_1(\alpha)}{K_2(\alpha)} = \frac{-2}{5}$.

Let $A \in \mathbb{LC}(A_1, A_2)$ with $A = tA_1 \oplus t^2A_2$ for all $t \in [0, 1]$. Then $q_1(t) = t$ and $r_1(t) = t^2$. We have

$$q_2(t) = K_{1,2}q(t) = \frac{-2}{5}t; \quad r_2(t) = 2q_1(t)l_{1,2} + r_1(t) = \frac{7}{10}t + t^2$$

Therefore, $\psi(t, t^2) = \psi\left(\frac{-2}{5}t, \frac{7}{10}t + t^2\right)$.

By Theorem 3.4, we have that

$$\psi(1, 2t) = \left(-1; \frac{1}{2} + 2t; \frac{5}{2} + 6t\right)$$

and

$$\psi\left(\frac{-2}{5}, \frac{7}{10} + 2t\right) = \left(-1; \frac{1}{2} + 2t; \frac{5}{2} + 6t\right).$$

Lastly, according to Theorem 3.3, we have $\psi(1, 2t) = \psi\left(\frac{-2}{5}, \frac{7}{10} + 2t\right)$.

Example 3.2. Let $g(t) = e^t A_1 \oplus (t^2 + 2)A_2$ where $A_1 = (-2; 0; 2)$ and $A_2 = (2; 3; 4)$. In this case, the pair $\{A_1, A_2\}$ is not SLI and hence, the Fréchet derivative is given by $g'_F(t) = e^t A_1 \oplus 2tA_2$.

3.2. The Riemann integral and fractional derivative

Let $A_1, A_2 \in \mathbb{R}_F$ and $f : J = [a, b] \rightarrow \mathbb{LC}(A_1, A_2)$ be a linearly correlated fuzzy number-valued function with $f(t) = \psi(q(t), r(t))$ for all $t \in [a, b]$.

Definition 3.2 ([16]). Let $\{A_1, A_2\}$ be a SLI set and $q, r : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on the closed interval $[a, b]$. Then, the Riemann integral of $f(t)$ on $[a, b]$ is defined by

$$\int_a^b f(t)dt = \left(\int_a^b q(t)dt \right) A_1 \oplus \left(\int_a^b r(t)dt \right) A_2, \quad t \in [a, b]. \quad (3)$$

Lemma 3.3. Let $\{A_1, A_2\}$ be a SLI set and $f : [a, b] \rightarrow \mathbb{L}\mathbb{C}(A_1, A_2)$ be Fréchet differentiable on (a, b) . If f'_F is Riemann integrable on $[a, b]$ then

$$\int_a^t f'_F(s)ds = f(t) \ominus f(a), \quad t \in [a, b].$$

Proof. Since f is Fréchet differentiable on (a, b) , there exist differentiable functions $q, r : [a, b] \rightarrow \mathbb{R}$ such that $f'_F(t) = \psi(q'(t), r'(t))$ for all $t \in [a, b]$. Thus, for all $t \in [a, b]$, we have

$$\begin{aligned} \int_a^t f'_F(s)ds &= \psi \left(\int_a^t q'(s)ds, \int_a^t r'(s)ds \right) \\ &= \psi(q(t) - q(a), r(t) - r(a)) \\ &= \psi(q(t), r(t)) \ominus \psi(q(a), r(a)) \\ &= f(t) \ominus f(a). \quad \square \end{aligned}$$

If $\{A_1, A_2\}$ is not SLI, based on Equation (3), its value may depend on the choice of the pair of functions (q, r) . The next theorem presents a condition for integrating fuzzy functions when $\{A_1, A_2\}$ is not SLI by means of Equation (3).

Theorem 3.5. Let A_1, A_2, A_3, A_4 be fuzzy numbers and two functions $\psi : \mathbb{R}^2 \rightarrow \mathbb{L}\mathbb{C}(A_1, A_2)$ and $\hat{\psi} : \mathbb{R}^2 \rightarrow \mathbb{L}\mathbb{C}(A_3, A_4)$. Assume that there exist Riemann-integrable functions $q_i, r_i : \mathbb{R} \rightarrow \mathbb{R}, i = 1, 2$, that do not change sign on $[a, b]$ such that

$$f(t) = \psi(q_1(t), r_1(t)) = \hat{\psi}(q_2(t), r_2(t)),$$

for all $t \in [a, b]$. Then we have

$$\psi \left(\int_a^b q_1(t)dt, \int_a^b r_1(t)dt \right) = \hat{\psi} \left(\int_a^b q_2(t)dt, \int_a^b r_2(t)dt \right), t \in [a, b].$$

Proof. Assume that A_1, A_2, A_3, A_4 are fuzzy numbers whose α -cuts are given by $[A_1]^\alpha = [A_{1\alpha}^-, A_{1\alpha}^+]$, $[A_2]^\alpha = [A_{2\alpha}^-, A_{2\alpha}^+]$, $[A_3]^\alpha = [A_{3\alpha}^-, A_{3\alpha}^+]$ and $[A_4]^\alpha = [A_{4\alpha}^-, A_{4\alpha}^+]$. Since q_i and r_i do not change sign on $[a, b]$, we consider the following cases:

- Case 1:** $q_1(t) \geq 0, r_1(t) \geq 0, q_2(t) \geq 0, r_2(t) \geq 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) < 0, r_2(t) < 0$.
- Case 2:** $q_1(t) \geq 0, r_1(t) \geq 0, q_2(t) < 0, r_2(t) \geq 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) \geq 0, r_2(t) < 0$.
- Case 3:** $q_1(t) \geq 0, r_1(t) \geq 0, q_2(t) \geq 0, r_2(t) < 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) \geq 0, r_2(t) \geq 0$.
- Case 4:** $q_1(t) \geq 0, r_1(t) \geq 0, q_2(t) < 0, r_2(t) < 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) \geq 0, r_2(t) \geq 0$.
- Case 5:** $q_1(t) \geq 0, r_1(t) < 0, q_2(t) \geq 0, r_2(t) < 0$ or $q_1(t) < 0, r_1(t) \geq 0, q_2(t) < 0, r_2(t) \geq 0$.
- Case 6:** $q_1(t) \geq 0, r_1(t) < 0, q_2(t) < 0, r_2(t) < 0$ or $q_1(t) < 0, r_1(t) \geq 0, q_2(t) \geq 0, r_2(t) \geq 0$.
- Case 7:** $q_1(t) \geq 0, r_1(t) < 0, q_2(t) \geq 0, r_2(t) \geq 0$ or $q_1(t) < 0, r_1(t) \geq 0, q_2(t) < 0, r_2(t) < 0$.
- Case 8:** $q_1(t) \geq 0, r_1(t) < 0, q_2(t) < 0, r_2(t) \geq 0$ or $q_1(t) < 0, r_1(t) \geq 0, q_2(t) \geq 0, r_2(t) < 0$.

Without loss of generality, we will prove the first case and the remaining cases can be proved similarly.

When $q_1(t) \geq 0, r_1(t) \geq 0, q_2(t) \geq 0, r_2(t) \geq 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) < 0, r_2(t) < 0$ for all $t \in [a, b]$. Since $q_1(t)A_1 \oplus r_1(t)A_2 = q_2(t)A_3 \oplus r_2(t)A_4$ for all $t \in [a, b]$ we have

$$\begin{cases} q_1(t)A_{1\alpha}^+ + r_1(t)A_{2\alpha}^+ = q_2(t)A_{3\alpha}^+ + r_2(t)A_{4\alpha}^+, \\ q_1(t)A_{1\alpha}^- + r_1(t)A_{2\alpha}^- = q_2(t)A_{3\alpha}^- + r_2(t)A_{4\alpha}^-, \end{cases} \quad t \in [a, b].$$

By integrating both sides, we directly obtain

$$\begin{cases} A_{1\alpha}^+ \int_a^b q_1(t)dt + A_{2\alpha}^+ \int_a^b r_1(t)dt = A_{3\alpha}^+ \int_a^b q_2(t)dt + A_{4\alpha}^+ \int_a^b q_1(t)dt \\ A_{1\alpha}^- \int_a^b q_1(t)dt + A_{2\alpha}^- \int_a^b r_1(t)dt = A_{3\alpha}^- \int_a^b q_2(t)dt + A_{4\alpha}^- \int_a^b r_2(t)dt, \end{cases} \quad t \in [a, b].$$

Since q_i and $\int_a^b q_i(t)dt$, r_i and $\int_a^b r_i(t)dt$ have the same sign for all $t \in [a, b]$. Therefore, we can conclude that

$$\psi \left(\int_a^b q_1(t)dt, \int_a^b r_1(t)dt \right) = \hat{\psi} \left(\int_a^b q_2(t)dt, \int_a^b r_2(t)dt \right), t \in [a, b].$$

This implies the conclusion. $\square$

From Theorem 3.5, we define the Riemann integral of the function $f(t)$ in the case when $\{A_1, A_2\}$ is not SLI.

Definition 3.3. Let $\{A_1, A_2\}$ be not SLI. Assume that $f(t) = q(t)A_1 + r(t)A_2$, where $q(t)$ and $r(t)$ are Riemann integrable on J and do not change sign on this interval. Then, the Riemann integral of the function $f(t)$ is defined by

$$\int_a^b f(t)dt = \psi \left(\int_a^b q(t)dt, \int_a^b r(t)dt \right), t \in J.$$

After giving the definition of the Riemann integral, we define the Riemann-Liouville fractional integral of the function $f(t) \in \mathbb{LC}(A_1, A_2)$.

Definition 3.4. Let A_1, A_2 be fuzzy numbers. Assume that $f(t) = q(t)A_1 + r(t)A_2$, where $q(t)$ and $r(t)$ are Riemann integrable on J .

1) If $\{A_1, A_2\}$ is SLI, the Riemann-Liouville fractional integral of order $p \in (0, 1]$ of the function f is defined by

$${}^{RL}I_{0+}^p f(t) = \psi \left(I_{0+}^p q(t), I_{0+}^p r(t) \right), t \in J,$$

2) If $\{A_1, A_2\}$ is not SLI and $q(t), r(t)$ do not change the sign on J , the Riemann-Liouville fractional integral of order $p \in (0, 1]$ of the function f is defined by

$${}^{RL}I_{0+}^p f(t) = \psi \left(I_{0+}^p q(t), I_{0+}^p r(t) \right), t \in J,$$

where $I_{0+}^p q(t)$ and $I_{0+}^p r(t)$ are the Riemann-Liouville fractional integrals of order $p \in (0, 1]$ of the functions $q(t), r(t)$ for all $t \in J$.

The validity of Definition 3.4 is established through Theorem 3.6. The theorem demonstrates that under the condition of non-sign-changing component functions on the domain J , the Riemann-Liouville fractional integral yields results independent of the chosen representation of the fuzzy function.

Theorem 3.6. Let A_1, A_2, A_3, A_4 be fuzzy numbers and two functions $\psi : \mathbb{R}^2 \rightarrow \mathbb{LC}(A_1, A_2)$ and $\hat{\psi} : \mathbb{R}^2 \rightarrow \mathbb{LC}(A_3, A_4)$. Assume that there exist Riemann-integrable functions $q_i, r_i : \mathbb{R} \rightarrow \mathbb{R}, i = 1, 2$, that do not change sign on $[a, b]$ such that

$$f(t) = \psi(q_1(t), r_1(t)) = \hat{\psi}(q_2(t), r_2(t))$$

for all $t \in J$. Then we have

$$\psi(I_{0+}^p q_1(t), I_{0+}^p r_1(t)) = \hat{\psi}(I_{0+}^p q_2(t), I_{0+}^p r_2(t)), t \in J. \quad (4)$$

Proof. Assume that A_1, A_2, A_3, A_4 are fuzzy numbers whose α -cuts are $[A_1]^\alpha = [A_{1\alpha}^-, A_{1\alpha}^+]$, $[A_2]^\alpha = [A_{2\alpha}^-, A_{2\alpha}^+]$, $[A_3]^\alpha = [A_{3\alpha}^-, A_{3\alpha}^+]$ and $[A_4]^\alpha = [A_{4\alpha}^-, A_{4\alpha}^+]$. Since $q_i(t)$ and $r_i(t)$ do not change of sign on J for all $i = 1, 2$, we have 8 cases as in Theorem 3.5. We will prove this theorem for the first case: $q_1(t) \geq 0, r_1(t) \geq 0, q_2(t) \geq 0, r_2(t) \geq 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) < 0, r_2(t) < 0$ for all $t \in J$. From $q_1(t)A_1 \oplus r_1(t)A_2 = q_2(t)A_3 \oplus r_2(t)A_4, t \in J$, we have

$$\begin{cases} q_1(t)A_{1\alpha}^+ + r_1(t)A_{2\alpha}^+ = q_2(t)A_{3\alpha}^+ + r_2(t)A_{4\alpha}^+, \\ q_1(t)A_{1\alpha}^- + r_1(t)A_{2\alpha}^- = q_2(t)A_{3\alpha}^- + r_2(t)A_{4\alpha}^-, \end{cases} \quad t \in J.$$

By taking fractional integral order $p \in (0, 1)$ for both sides of this system, we obtain

$$\begin{cases} I_{0+}^p q_1(t) A_{1\alpha}^+ + I_{0+}^p r_1(t) A_{2\alpha}^+ = I_{0+}^p q_2(t) A_{3\alpha}^+ + I_{0+}^p r_2(t) A_{4\alpha}^+, \\ I_{0+}^p q_1(t) A_{1\alpha}^- + I_{0+}^p r_1(t) A_{2\alpha}^- = I_{0+}^p q_2(t) A_{3\alpha}^- + I_{0+}^p r_2(t) A_{4\alpha}^-, \end{cases} \quad t \in J.$$

Since $q_1(t) \geq 0, r_1(t) \geq 0, q_2(t) \geq 0, r_2(t) \geq 0$ or $q_1(t) < 0, r_1(t) < 0, q_2(t) < 0, r_2(t) < 0$ for all $t \in J$ then $I_{0+}^p q_1(t) \geq 0, I_{0+}^p r_1(t) \geq 0, I_{0+}^p q_2(t) \geq 0, I_{0+}^p r_2(t) \geq 0$ or $I_{0+}^p q_1(t) < 0, I_{0+}^p r_1(t) < 0, I_{0+}^p q_2(t) < 0, I_{0+}^p r_2(t) < 0$ for all $t \in J$. We can conclude that

$$\psi(I_{0+}^p q_1(t), I_{0+}^p r_1(t)) = \psi(I_{0+}^p q_2(t), I_{0+}^p r_2(t)), \quad t \in J.$$

The remaining cases can be proved similarly. $\square$

Definition 3.5. Let $\{A_1, A_2\}$ be a SLI set and $q(t), r(t)$ be continuously differentiable function on J . For $p \in (0, 1]$, the Fréchet Caputo fractional derivative of order p of the function $f(t)$ is defined by

$${}_F^C D_{0+}^p f(t) = {}_F^{RL} I_{0+}^p f'(t) = \psi \left(I_{0+}^{1-p} q'(t), I_{0+}^{1-p} r'(t) \right), \quad t \in J. \quad (5)$$

In the case when $\{A_1, A_2\}$ is not SLI, let us consider a function $f(t) = q(t)A_1 \oplus r(t)A_2$ such that $q'(\cdot)$ and $r'(\cdot)$ do not change sign on J . Then, we define the Fréchet Caputo fractional derivative of order p of the function f by (5).

Next, we will introduce some important results on the Laplace transform of a linearly correlated fuzzy number-valued function $f : J \rightarrow \mathbb{LC}(A_1, A_2)$.

Definition 3.6. Let A_1, A_2 be fuzzy numbers and $f : J \rightarrow \mathbb{LC}(A_1, A_2)$ be a linearly correlated fuzzy number-valued function. Assume that the function $f(t)$ is given by $f(t) = \psi(q(t), r(t))$, where $q(t)$ and $r(t)$ are continuous real-valued functions. Then, the fuzzy Laplace transform of the function $f(t)$ is given by

$$\mathcal{L}[f(t)](s) = \psi(L[q(t)](s), L[r(t)](s)), \quad t \in J, s \in [0, \infty), p \in (0, 1],$$

where $L[q(t)](s)$ and $L[r(t)](s)$ are classical Laplace transforms of real functions $q(t)$ and $r(t)$ for all $t \in J$ and $s \in [0, \infty)$.

Proposition 3.1. Assume that $\{A_1, A_2\}$ is a SLI set and $f : J \rightarrow \mathbb{LC}(A_1, A_2)$ is Fréchet Caputo fractional differentiable on $[0, \infty)$. Then,

$$\mathcal{L}({}_F^C D_{0+}^p f(t))(s) = s^p \cdot \mathcal{L}[f(t)](z) \ominus s^{p-1} \cdot z_0, \quad t \in J, s \in [0, \infty). \quad (6)$$

Proof. Indeed, for each $t \in J$, we have

$${}_F^C D_{0+}^p f(t) = \psi \left({}^C D_{0+}^p q(t), {}^C D_{0+}^p r(t) \right).$$

It follows that for $t \in J, z \in [0, \infty)$, we have

$$\mathcal{L}({}_F^C D_{0+}^p f(t))(s) = \psi(L[{}^C D_{0+}^p q(t)](s), L[{}^C D_{0+}^p r(t)](s)).$$

According to the properties of Laplace transform of real functions, we obtain

$$\begin{aligned} \mathcal{L}({}_F^C D_{0+}^p f(t))(s) &= \psi(z^p L[q(t)](s) - s^{p-1} q(0), s^p L[r(t)](s) - s^{p-1} r(0)) \\ &= (s^p L[q(t)](s) - s^{p-1} q(0)A_1 + (s^p L[r(t)](s) - s^{p-1} r(0)A_2) \\ &= s^p (L[q(t)](s)A_1 + L[r(t)](s)A_2) \ominus s^{p-1} (q(0)A_1 + r(0)A_2) \\ &= s^p \mathcal{L}[x(t)](s) \ominus s^{p-1} x_0. \end{aligned}$$

This completes the proof. $\square$

Proposition 3.2. Let $\{A_1, A_2\}$ be a SLI set and $k : J \rightarrow \mathbb{R}, f : J \rightarrow \mathbb{LC}(A_1, A_2)$ be two continuous functions. Then, we have

$$\mathcal{L}[k(t) * f(t)](s) = L[k(t)](s) \mathcal{L}[f(t)](z), \quad (7)$$

where $k(t) * f(t) = \int_0^t k(t-\tau)f(\tau)d\tau, t \in J$ and $s \in [0, \infty)$.

Proof. Since $\{A_1, A_2\}$ is SLI, $k(t) * f(t) = \psi(k(t) * q(t), k(t) * r(t))$. This leads to the result that for all $t \in J, s \in (0, t)$, we get

$$\mathcal{L}[k(t) * f(t)](s) = \mathcal{L}[\psi(k(t) * q(t), k(t) * r(t))](s)$$

$$\begin{aligned}
 &= \psi(L[k(t) * q(t)](s), L[k(t) * r(t)](s)) \\
 &= \psi(L[k(t)](s)L[q(t)](s), L[k(t)](s)L[r(t)](s)) \\
 &= L[k(t)](s)\psi(L[q(t)](s), L[r(t)](s)) \\
 &= L[k(t)](s)\mathcal{L}[x(t)](s).
 \end{aligned}$$

Therefore, the proof is completed. $\square$

4. The observability of fuzzy fractional evolution equations in $[\mathbb{L}\mathbb{C}(A_1, A_2)]^n$

Let $[\mathbb{L}\mathbb{C}(A_1, A_2)]^n := \{z = (z_1, z_2, \dots, z_n)^\top : z_i \in \mathbb{L}\mathbb{C}(A_1, A_2), i = \overline{1, n}\}$ be a set consisting of all fuzzy vectors whose components take values in $\mathbb{L}\mathbb{C}(A_1, A_2)$. In this section, we aim to study the observability of a class of fuzzy fractional linear evolution equations in the space $[\mathbb{L}\mathbb{C}(A_1, A_2)]^n$ in both cases: $\{A_1, A_2\}$ is SLI and $\{A_1, A_2\}$ is not SLI.

4.1. The case when $\{A_1, A_2\}$ is SLI

In the case where the pair $\{A_1, A_2\}$ is SLI, we define a metric on the space $[\mathbb{L}\mathbb{C}(A_1, A_2)]^n$ by the mapping $d_{\mathbb{L}\mathbb{C}} : [\mathbb{L}\mathbb{C}(A_1, A_2)]^n \times [\mathbb{L}\mathbb{C}(A_1, A_2)]^n \rightarrow \mathbb{R}$

$$d_{\mathbb{L}\mathbb{C}}(\mathbf{u}, \mathbf{v}) = \sqrt{d^2(u_1, v_1) + d^2(u_2, v_2) + \dots + d^2(u_n, v_n)},$$

for all $\mathbf{u} = (u_1, \dots, u_n)^\top, \mathbf{v} = (v_1, \dots, v_n)^\top \in [\mathbb{L}\mathbb{C}(A_1, A_2)]^n$. Since the metric space $([\mathbb{L}\mathbb{C}(A_1, A_2)], d)$ is complete, we can easily verify that the metric space $([\mathbb{L}\mathbb{C}(A_1, A_2)], d_{\mathbb{L}\mathbb{C}})$ is also complete.

Remark 4.1. It is obvious that for each $\mathbf{u}, \mathbf{v}, \mathbf{w} \in [\mathbb{L}\mathbb{C}(A_1, A_2)]^n$, the following equality holds

$$d_{\mathbb{L}\mathbb{C}}(\mathbf{u} \oplus \mathbf{w}, \mathbf{v} \oplus \mathbf{w}) = d_{\mathbb{L}\mathbb{C}}(\mathbf{u}, \mathbf{v}).$$

Denote by $C(J, [\mathbb{L}\mathbb{C}(A_1, A_2)]^n)$ the space of all fuzzy-valued functions $\mathbf{x} : J \rightarrow [\mathbb{L}\mathbb{C}(A_1, A_2)]^n$ that are continuous on $J = [0, T]$ with $T > 0$. This functional space is a complete metric space endowed with the supremum metric

$$\rho_{\mathbb{L}\mathbb{C}}^n(\mathbf{u}, \mathbf{v}) = \sup_{t \in J} d_{\mathbb{L}\mathbb{C}}(\mathbf{u}(t), \mathbf{v}(t)),$$

for all $\mathbf{u}, \mathbf{v} \in C(J, [\mathbb{L}\mathbb{C}(A_1, A_2)]^n)$.

In this subsection, we investigate the qualitative properties of the following initial value problem for fuzzy fractional linear evolution equations in the case when $\{A_1, A_2\}$ is SLI:

$${}_J^C D_{0+}^p \mathbf{x}(t) = B\mathbf{x}(t) \tag{8}$$

subject to the initial condition

$$\mathbf{x}(0) = \mathbf{x}_0, \tag{9}$$

where ${}_J^C D_{0+}^p \mathbf{x}(t)$ is the Caputo fractional derivative operator of order $p \in (0, 1)$ of the state function $\mathbf{x}(t) \in [\mathbb{L}\mathbb{C}(A_1, A_2)]^n$, B is a given square matrix of order n and $\mathbf{x}_0 \in [\mathbb{L}\mathbb{C}(A_1, A_2)]^n$. Moreover, consider a linear observation

$$\mathbf{y}(t) = H\mathbf{x}(t), \tag{10}$$

where $\mathbf{y}(t) \in [\mathbb{L}\mathbb{C}(A_1, A_2)]^m$ is a fuzzy vector function, namely the output function, and H is a given matrix of order $m \times n$. The main aim of this section is to find sufficient conditions such that the fractional linear evolution equations (8) with the linear observation (10) are completely observable on $[0, T]$. Based on the preceding concept mentioned in [22], we introduce the concept of observability for the system (8) and (10):

Definition 4.1. The fractional linear evolution equations (8) with the observation (10) are called observable at time t_f with $t_f \in (0, T]$ if an arbitrary initial state $\mathbf{x}_0$ can be determined uniquely from the output function $\mathbf{y}(t)$ for all $t \in [0, t_f]$. If this system is observable at every $t_f \in [0, T]$ then we say that the system (8) and (10) is completely observable on $[0, T]$.

Next, we define the observability Gramian matrix function $W_0 : [0, t_f] \rightarrow \mathbb{R}$ by

$$W_0(t) = \int_0^t \mathbb{E}_p(B^\top t^p) H^\top H \mathbb{E}_p(B t^p) dt, \quad t \in [0, t_f], \tag{11}$$

where $B^\top$ and $H^\top$ denote for transpose matrices of the matrices B and H , respectively. In addition, since the fact that

$$\begin{aligned} W_0^\top(t) &= \left(\int_0^t \mathbb{E}_p(B^\top t^p) H^\top H \mathbb{E}_p(B t^p) dt \right)^\top \\ &= \int_0^t (\mathbb{E}_p(B^\top t^p) H^\top H \mathbb{E}_p(B t^p))^\top dt \\ &= \int_0^t \mathbb{E}_p(B^\top t^p) H^\top H \mathbb{E}_p(B t^p) dt \\ &= W_0(t), \end{aligned}$$

for all $t \in [0, t_f]$, we can conclude that the matrix $W_0(t)$ is symmetric. Furthermore, the function $\mathbb{E}_p(Bt)$ is known as the two-parameter Mittag-Leffler matrix function defined by

$$\mathbb{E}_{p_1, p_2}(Bt) = \sum_{k=0}^{\infty} \frac{t^k}{\Gamma(p_1 k + p_2)} B^k.$$

In the case when $p_2 = 1$, we denote $\mathbb{E}_{p_1, 1}(Bt) = \mathbb{E}_{p_1}(Bt)$. Some useful properties of the two-parameter Mittag-Leffler matrix function can be given as follows:

Lemma 4.1. Let $B = [b_{ij}]_{n \times n}$ be a real matrix. Then, for each $p_1, p_2 > 0$, the following properties are fulfilled

(i) If B is a diagonal matrix, i.e., $B = \text{diag}(b_{11}, b_{22}, \dots, b_{nn})$, then

$$\mathbb{E}_{p_1, p_2}(B) = \text{diag}(\mathbb{E}_{p_1, p_2}(b_{11}), \mathbb{E}_{p_1, p_2}(b_{22}), \dots, \mathbb{E}_{p_1, p_2}(b_{nn})).$$

(ii) If there exists an invertible matrix S and a diagonal matrix D such that $B = SDS^{-1}$ then

$$\mathbb{E}_{p_1, p_2}(B) = S \mathbb{E}_{p_1, p_2}(D) S^{-1}.$$

(iii) $\mathbb{E}_{p_1, p_2}(B^\top) = [\mathbb{E}_{p_1, p_2}(B)]^\top$.

(iv) The norm of the matrix $\mathbb{E}_{p_1, p_2}(B)$ satisfies $\|\mathbb{E}_{p_1, p_2}(B)\| \leq \sum_{k=0}^{\infty} \frac{\|B\|^k}{\Gamma(p_1 k + p_2)} < \infty$.

(v) (see [28]) For each $p_1, p_2 > 0$ and $t \geq 0$, the following Laplace transform holds

$$L \left[t^{p_2-1} \mathbb{E}_{p_1, p_2}(B t^{p_1}) \right] (s) = s^{p_1-p_2} (s^{p_1} \mathbb{I}_n - B)^{-1},$$

for all $s > \|B\|^{\frac{1}{p_1}}$ and $\mathbb{I}_n$ is the identity matrix.

Proposition 4.1. The initial value problem (8)-(9) for fractional linear differential systems has a unique fuzzy solution of the form

$$\mathbf{x}(t) = \mathbb{E}_p(B t^p) \mathbf{x}_0, \quad t \in J.$$

Proof. For each $t \in J$ and $s > \|B\|^{\frac{1}{p}}$, denote $X(s) := \mathcal{L}[\mathbf{x}(t)](s)$. Then, by taking the Laplace transform for both sides of the fractional differential system (8), we have

$$s^p X(s) \ominus s^{p-1} \mathbf{x}_0 = B X(s)$$

or equivalently, $X(s) = \frac{s^{p-1}}{s^p \mathbb{I}_n - B} \mathbf{x}_0$. By using Lemma 4.1-(v), we obtain

$$X(s) = L \left[\mathbb{E}_{p, 1}(B t^p) \right] (s) \mathbf{x}_0. \quad (12)$$

Finally, by taking the inverse Laplace transform on both sides of (12), we immediately get

$$\mathbf{x}(t) = \mathbb{E}_p(B t^p) \mathbf{x}_0, \quad t \in J.$$

Therefore, the proof is completed. $\square$

Now, we turn to one of the main results of this section

Theorem 4.1. *If the matrix $W_0(t)$ is non-singular for all $t \in [0, t_f]$ then the fractional linear evolution equations (8) with the observation (10) are observable at time t_f .*

Proof. For each $t \in [0, t_f]$, it follows from Proposition 4.1 that $\mathbf{x}(t) = \mathbb{E}_p(Bt^p)\mathbf{x}_0$. Thus, it implies that the output function $\mathbf{y}(t)$ is given by

$$\mathbf{y}(t) = H\mathbb{E}_p(Bt^p)\mathbf{x}_0.$$

The observability of the system (8)-(10) is equivalent to the estimation of $\mathbf{x}_0$ from $\mathbf{y}(t)$. Since the matrix $W_0(t_f)$ is non-singular then the matrix $W_0^{-1}(z(t_f))$ is well-defined. Therefore, for any arbitrary output function $\mathbf{y}(t)$ and $t_f > 0$, by multiplying both sides of the above equation by $\mathbb{E}_p(B^\top t^p)H^\top$ from the left and integrating from 0 to t_f , we obtain

$$\begin{aligned} W_0^{-1}(z(t_f)) \int_0^{t_f} \mathbb{E}_p(B^\top t^p)H^\top \mathbf{y}(t)dt &= W_0^{-1}(z(t_f)) \int_0^{t_f} \mathbb{E}_p(B^\top t^p)H^\top H\mathbb{E}_p(Bt^p)\mathbf{x}_0 dt \\ &= W_0^{-1}(z(t_f)) W_0(t_f)\mathbf{x}_0 \\ &= \mathbf{x}_0. \end{aligned}$$

Thus, we obtain $\mathbf{x}_0 = W_0^{-1}(z(t_f)) \int_0^{t_f} \mathbb{E}_p(B^\top t^p)H^\top \mathbf{y}(t)dt$ and this equation can be seen as a linear algebraic equation of $\mathbf{x}_0$. Since the matrix $W_0(t_f)$ is non-singular then we can conclude that the initial value $\mathbf{x}_0$ can be uniquely determined by the corresponding output function $\mathbf{y}(t)$, $t \in [0, t_f]$. Therefore, the proof is completed. $\square$

4.2. The case when $\{A_1, A_2\}$ is not SLI

According to the results presented in Section 2, we have 38 cases of $\psi^{-1}(\psi(q, r))$ corresponding to the cases of A_1, A_2 . We can summarize some particular cases in which the pair of fuzzy numbers $\{A_1, A_2\}$ is not SLI including the case when A_1, A_2 are non-symmetric fuzzy numbers and $\rho_{A_2}(\alpha) = k\rho_{A_1}(\alpha)$ with $k \in \mathbb{R} \setminus \{0\}$, the case when A_1 is non-symmetric fuzzy number and A_2 is a symmetric fuzzy number with the point $x^* = 0$ or the case when A_1, A_2 are all symmetric fuzzy numbers. For these above cases, our main idea is to construct the formulas of the pre-image mapping $\psi^{-1}(A)$. Indeed, in Theorem 2.3, Theorem 2.1, Theorem 2.2, Theorem 2.4, Theorem 2.5 and Theorem 2.6, we find out detailed conditions for which the inverse image of $\psi(q, r)$ exists and give explicit formulas for the mapping $\psi^{-1}(\psi(q, r))$. It should be noted that since the mapping ψ is not a bijection in the case when $\{A_1, A_2\}$ is not SLI and the mapping $\psi^{-1}(\psi(q, r))$ has more than one pre-image, we cannot construct any isomorphism between $\mathbb{R}^2$ and $\psi(q, r)$. As a consequence, the space $\mathbb{LC}(A_1, A_2)$ is not a linear (or vector) space and hence, it is not a Banach space. This leads to a big challenge in the study of fuzzy-valued functions and fuzzy dynamical systems taking values in $\mathbb{LC}(A_1, A_2)$. Based on the idea introduced by Y. Shen [31], we will construct an equivalence relation $\equiv_{\mathbb{LC}}$ on $\mathbb{R}^2$ and consider a closed quotient space of $\mathbb{R}^2$, denoted by $\mathbb{R}^2 / \equiv_{\mathbb{LC}}$, based on the equivalence class $[(q, r)]_{\equiv_{\mathbb{LC}}}$ generated by the above equivalence relation. After that, we facilitate some algebraic operations on the space $\mathbb{R}^2 / \equiv_{\mathbb{LC}}$ consisting of addition $\oplus$, scalar multiplication and difference $\ominus$. Especially, the mapping $\hat{\psi} : \mathbb{R}^2 / \equiv_{\mathbb{LC}} \rightarrow \mathbb{LC}(A_1, A_2)$, given by $\hat{\psi}([(q, r)]_{\equiv_{\mathbb{LC}}}) = qA_1 \oplus rA_2$, is now a bijection and the algebraic operations on the space $\mathbb{LC}(A_1, A_2)$ are then defined via the mapping $\hat{\psi}$. Hence, this approach provides an efficient solution to overcome the difficulty in dealing with the space $\mathbb{LC}(A_1, A_2)$ when the pair $\{A_1, A_2\}$ is not SLI. By using the results of theorems (from Theorem 2.3 to Theorem 2.6), we can construct many quotient spaces of $\mathbb{R}^2$. Without loss of generality, this subsection only considers a particular case of a not SLI pair of fuzzy numbers $\{A_1, A_2\}$, that is, A_1 and A_2 are non-symmetric fuzzy numbers and $A_2 = kA_1$ with the parameter $k > 0$ and $K_1(\alpha) = \text{len}[A_2](\alpha) - k\text{len}[A_1](\alpha) \leq 0$ for all $\alpha \in [0, 1]$. Consider the space $\mathbb{LC}(A_1, A_2)$ consisting of all fuzzy numbers $A = qA_1 \oplus rA_2$ with $q, r \geq 0$. According to Theorem 2.1, we can present the pre-image ψ^{-1} as follows:

$$\psi^{-1}(A) = \{(q, r), (0, 2ql_{1,2} + r)\}$$

for all $A \in \mathbb{LC}(A_1, A_2)$ with $q, r \geq 0$ and $l_{1,2} = \frac{\text{len}[A_1](\alpha)}{K_2(\alpha)}$.

For simplicity in representation, we denote by

$$\mathbf{X}_1 = \{A = qA_1 \oplus rA_2 \in \mathbb{LC}(A_1, A_2) : q, r \geq 0, \psi^{-1}(A) = \{(q, r), (0, 2ql_{1,2} + r)\}\}$$

the set of all fuzzy numbers $A \in \mathbb{LC}(A_1, A_2)$ satisfying the above particular case. In order to introduce the algebraic operations in $\mathbb{LC}(A_1, A_2)$, we define an equivalence relation $\equiv_{\mathbb{LC}}$ by

$$(q, r) \equiv_{\mathbb{L}C} (p, s) \Leftrightarrow (q, r) = (p, s) \text{ or } (p, s) = (0, 2ql_{1,2} + r),$$

and an equivalence class $[(q, r)]_{\equiv_{\mathbb{L}C}} := \{(q, r), (0, 2ql_{1,2} + r)\}$. Next, by using the equivalence relation $\equiv_{\mathbb{L}C}$, the quotient set of $\mathbb{R}^2$ is defined by

$$\mathbb{R}^2 / \equiv_{\mathbb{L}C} := \{[(q, r)]_{\equiv_{\mathbb{L}C}} : (q, r) \in \mathbb{R}^2\}.$$

Now, we facilitate the algebraic operations in $\mathbb{R}^2 / \equiv_{\mathbb{L}C}$ as follows:

$$\begin{aligned} [(q, r)]_{\equiv_{\mathbb{L}C}} \hat{\oplus} [(p, s)]_{\equiv_{\mathbb{L}C}} &= [(q + p, r + s)]_{\equiv_{\mathbb{L}C}} \\ \lambda [(q, r)]_{\equiv_{\mathbb{L}C}} &= \begin{cases} [(\lambda q, \lambda r)]_{\equiv_{\mathbb{L}C}} & \text{if } \lambda \geq 0, \\ [(0, 2\lambda ql_{1,2} + \lambda r)]_{\equiv_{\mathbb{L}C}} & \text{if } \lambda < 0, \end{cases} \\ [(q, r)]_{\equiv_{\mathbb{L}C}} \hat{\ominus} [(p, s)]_{\equiv_{\mathbb{L}C}} &= \begin{cases} [(q - p, r - s)]_{\equiv_{\mathbb{L}C}} & \text{if } q \geq p, \\ [(0, 2(p - q)l_{1,2} + r - s)]_{\equiv_{\mathbb{L}C}} & \text{if } q < p, \end{cases} \end{aligned}$$

for all $[(q, r)]_{\equiv_{\mathbb{L}C}}, [(p, s)]_{\equiv_{\mathbb{L}C}} \in \mathbb{R}^2 / \equiv_{\mathbb{L}C}$ and $\lambda \in \mathbb{R}$. Next, we introduce an operator $\hat{\psi} : \mathbb{R}^2 / \equiv_{\mathbb{L}C} \rightarrow \mathbf{X}_1$ defined by

$$\hat{\psi}([(q, r)]_{\equiv_{\mathbb{L}C}}) = qA_1 \oplus rA_2,$$

for all $[(q, r)]_{\equiv_{\mathbb{L}C}} \in \mathbb{R}^2 / \equiv_{\mathbb{L}C}$. Clearly, the mapping $\hat{\psi}$ is a bijection. For each $B = \hat{\psi}([(q, r)]_{\equiv_{\mathbb{L}C}})$ and $C = \hat{\psi}([(p, s)]_{\equiv_{\mathbb{L}C}})$, we can define the operations on $\mathbf{X}_1$ based on $\hat{\psi}$ as follows:

$$\begin{aligned} B \hat{\oplus} C &= \hat{\psi}(\hat{\psi}^{-1}(B) + \hat{\psi}^{-1}(C)) \\ \lambda B &= \hat{\psi}(\lambda \hat{\psi}^{-1}(B)) \\ B \hat{\ominus} C &= \hat{\psi}(\hat{\psi}^{-1}(B) - \hat{\psi}^{-1}(C)). \end{aligned}$$

Lemma 4.2. Let B, C and $D \in \mathbf{X}_1$. Then, we have

- (i) $B \hat{\oplus} C = D$ implies $B = D \hat{\ominus} C$ and $C = D \hat{\ominus} B$.
- (ii) $B = D \hat{\ominus} C$ implies $B \hat{\oplus} C = D$ or $((-1)B) \hat{\oplus} D = C$.

Proof. The proof is straightforward from the definition of the equivalent class $[(q, r)]_{\equiv_{\mathbb{L}C}}$ and the algebraic operations on $\mathbf{X}_1$. $\square$

For each $[(q, r)]_{\equiv_{\mathbb{L}C}} \in \mathbb{R}^2 / \equiv_{\mathbb{L}C}$, we define the norm in the quotient space $\mathbb{R}^2 / \equiv_{\mathbb{L}C}$ by

$$\begin{aligned} \|[(q, r)]_{\equiv_{\mathbb{L}C}}\|_{\infty} &= \max \{ \|(q, r)\|_{\infty}, \|(qK_{1,2}, 2ql_{1,2} + r)\|_{\infty} \} \\ &= \max \{ |q|, |r|, |qK_{1,2}|, |2ql_{1,2} + r| \}. \end{aligned}$$

Then, for each $A \in \mathbf{X}_1$, the norm on $\mathbf{X}_1$ is given by

$$\|A\|_* = \|\hat{\psi}^{-1}(A)\|_{\infty}$$

and the metric d_* induced by this norm is $d_*(A, B) = \|A \hat{\ominus} B\|_*$. Moreover, since the mapping $\hat{\psi}$ is a bijection from $\mathbf{X}_1$ to $\mathbb{R}^2$, it implies that the space $(\mathbf{X}_1, d_*(\cdot, \cdot))$ is a Banach space.

Next, we define the space $\mathbf{X}_1^n := \{z = (z_1, \dots, z_n)^{\top} : z_i \in \mathbf{X}_1, i = \overline{1, n}\}$ by a set consisting of all fuzzy vectors whose components take values in $\mathbf{X}_1$ and a functional space $C(J, \mathbf{X}_1^n)$ consisting of all continuous functions $\mathbf{x} : J \rightarrow \mathbf{X}_1$. Since the completeness of $\mathbf{X}_1$, it implies that the space $C(J, \mathbf{X}_1^n)$ is also a complete metric space with the supremum metric

$$\rho_{\mathbf{X}_1^n}^n(\mathbf{u}, \mathbf{v}) = \sup_{t \in J} d_{\mathbf{X}_1}(\mathbf{u}(t), \mathbf{v}(t)),$$

for all $\mathbf{u}, \mathbf{v} \in C(J, \mathbf{X}_1^n)$, where the mapping $d_{\mathbf{X}_1} : \mathbf{X}_1 \times \mathbf{X}_1 \rightarrow \mathbb{R}$ is given by

$$d_{\mathbf{X}_1}(\mathbf{u}, \mathbf{v}) = \sqrt{d_*^2(u_1, v_1) + d_*^2(u_2, v_2) + \dots + d_*^2(u_n, v_n)},$$

for all $\mathbf{a} = (a_1, \dots, a_n)^{\top}, \mathbf{b} = (b_1, \dots, b_n)^{\top} \in \mathbf{X}_1^n$.

In this subsection, we aim to study the observability of a fuzzy fractional linear differential equations when the pair $\{A_1, A_2\}$ is not SLI under the conditions of Case 1.1. Indeed, we consider the following fractional linear differential system

$${}_F^C D_{0+}^p \mathbf{x}(t) = B\mathbf{x}(t) \tag{13}$$

with the linear observation

$$\mathbf{y}(t) = H\mathbf{x}(t), \quad (14)$$

and the initial condition

$$\mathbf{x}(0) = \mathbf{x}_0, \quad (15)$$

where ${}^C D_{0+}^p \mathbf{x}(t)$ is the Caputo fractional derivative operator of order $p \in (0, 1)$ of the state function $\mathbf{x}(t) \in \mathbf{X}_1^n$, $\mathbf{x}_0 \in \mathbf{X}_1^n$ and $\mathbf{y}(t) \in \mathbf{X}_1^m$ is the output function. The matrices B , H are given real matrices of order $n \times n$ and $m \times n$, respectively. Our aim is to find sufficient conditions such that the fractional linear differential equations (13) with the linear observation (14) are observable at time $t_f \in (0, T]$. Firstly, we recall the observability Gramian matrix $W_0(t)$ in (11) and by doing similar arguments, we obtain the following results:

Proposition 4.2. *The initial value problem (13)-(15) for the fractional linear differential system (13) has a unique fuzzy solution $\mathbf{x} \in C(J, \mathbf{X}_1^n)$ of the form*

$$\mathbf{x}(t) = \mathbb{E}_p(Bt^p)\mathbf{x}_0, \quad t \in J.$$

Proof. For each $t \in J$ and $s > \|B\|^{\frac{1}{p}}$, denote $X(s) := \mathcal{L}[\mathbf{x}(t)](s)$. Then, by taking the Laplace transform for both sides of the fractional differential system (8), we have

$$s^p X(s) \boxminus s^{p-1} \mathbf{x}_0 = BX(s).$$

According to Lemma 4.2, the above equality implies that

$$s^p X(s) = s^{p-1} \mathbf{x}_0 \hat{\oplus} BX(s) \text{ or } (-1)BX(s) \hat{\oplus} s^p X(s) = s^{p-1} \mathbf{x}_0.$$

If $s^p X(s) = s^{p-1} \mathbf{x}_0 \hat{\oplus} BX(s)$, by using the definition of the difference $\boxminus$ and $s > \|B\|^{\frac{1}{p}}$, we imply that $s^p X(s) \boxminus BX(s) = (s^p \mathbf{I}_n - B)X(s) = s^{p-1} \mathbf{x}_0$, or equivalently, $X(s) = s^{p-1}(s^p \mathbf{I}_n - B)^{-1} \mathbf{x}_0$.

In the case when $(-1)BX(s) \hat{\oplus} s^p X(s) = s^{p-1} \mathbf{x}_0$, by using the definition of the addition $\hat{\oplus}$, we have

$$\begin{aligned} (-1)BX(s) \hat{\oplus} s^p X(s) &= \hat{\psi}((-1)B\hat{\psi}^{-1}(X(s)) + s^p \mathbf{I}_n \hat{\psi}^{-1}(X(s))) \\ &= \hat{\psi}((s^p \mathbf{I}_n - B)\hat{\psi}^{-1}(X(s))). \end{aligned}$$

Then, we also obtain $X(s) = s^{p-1}(s^p \mathbf{I}_n - B)^{-1} \mathbf{x}_0$. By using Lemma 4.1-(v), we have

$$X(s) = L[\mathbb{E}_{p,1}(Bt^p)](s)\mathbf{x}_0. \quad (16)$$

Finally, by taking the inverse Laplace transform for both sides of (16), we immediately get

$$\mathbf{x}(t) = \mathbb{E}_p(Bt^p)\mathbf{x}_0, \quad t \in J.$$

Therefore, the proof is completed. $\square$

Theorem 4.2. *If the matrix $W_0(t)$ is non-singular for all $t \in [0, t_f]$ then the fractional linear evolution equations (13) with the linear observation (14) are observable at time t_f .*

Proof. By using similar arguments as in Theorem 4.1, the proof is straightforward. $\square$

Remark 4.2. In this section, we only consider one particular case when A_1, A_2 are non-symmetric fuzzy numbers and $A_2 = kA_1$, which follows that the pair $\{A_1, A_2\}$ is not SLI. Based on the formula of the pre-image mapping ψ^{-1} , we can construct an equivalence relation $\equiv_{\mathbb{L}\mathbb{C}}$ and then, consider a quotient space of $\mathbb{R}^2$ based on this relation. As a result, we obtain an one-to-one correspondence between the fuzzy space $\mathbf{X}_1$ and $\mathbb{R}^2 / \equiv_{\mathbb{L}\mathbb{C}}$. This approach helps us preserve many good properties of the quotient space of $\mathbb{R}^2$ and construct a Banach space of linearly correlated fuzzy numbers when $\{A_1, A_2\}$ is not SLI. It opens up many useful applications in studying dynamical systems that takes values in $\mathbb{L}\mathbb{C}(A_1, A_2)$ with $\{A_1, A_2\}$ is not SLI. Some other particular cases when the pair $\{A_1, A_2\}$ is not SLI may be considered similarly.

Example 4.1. In this example, we consider a fuzzy fractional linear evolution equations governed by the following fractional differential system

$$\begin{cases} {}^C D_{0+}^{\frac{1}{2}} x_1(t) = x_2(t) \\ {}^C D_{0+}^{\frac{1}{2}} x_2(t) = (-1)x_1(t), \end{cases} \quad (17)$$

with fuzzy initial condition

$$x_1(0) = A, \quad x_2(0) = \tilde{A}$$

and a linear observation:

$$y(t) = Hx(t), \quad (19)$$

where ${}^C_r D_{0+}^p x(t)$ denotes for fuzzy fractional derivative in Caputo sense of order $p = \frac{1}{2}$ of the state function $x(t) = [x_1(t) \ x_2(t)]^\top \in (\mathbb{LC}(A_1, A_2))^2$ for all $t \in J = [0, 10]$ and the matrix $H = \begin{bmatrix} 1 & 0 \end{bmatrix}$.

Denote the initial condition $x(0)$ by $x_0 = [A \ \tilde{A}]^\top \in (\mathbb{LC}(A_1, A_2))^2$ and let

$$B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Then, we can rewrite the initial value problem for the fractional differential system (17) with the initial condition (18) as follows:

$$\begin{cases} {}^C_r D_{0+}^{\frac{1}{2}} x(t) = Bx(t), & t \in J, \\ x(0) = x_0. \end{cases} \quad (20)$$

The initial value problem (20) has a unique solution of the form

$$x(t) = \mathbb{E}_{\frac{1}{2}}(Bt^{\frac{1}{2}})x_0, \quad t \in J,$$

where the Mittag-Leffler matrix function for the square matrix B is given by

$$\mathbb{E}_{\frac{1}{2}}(Bt^{\frac{1}{2}}) = \begin{bmatrix} e^{-t} & \sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t) \\ -\sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t) & e^{-t} \end{bmatrix}.$$

Since $A, \tilde{A} \in \mathbb{LC}(A_1, A_2)$, we consider two cases of the pair $\{A_1, A_2\}$ as follows:

Case 1: Let $A_1 = (1; 3; 4)$ and $A_2 = (-1; 2; 4)$ be two triangular fuzzy numbers. Since the functions $\rho_{A_1}(\alpha) = 5 + \alpha$ and $\rho_{A_2} = 3 + \alpha$ are linearly independent then the pair of fuzzy numbers $\{A_1, A_2\}$ is SLI. Hence, the space $[\mathbb{LC}(A_1, A_2)]^2$ is a complete metric space endowed with the metric

$$d_{\mathbb{LC}}(u, v) = \sqrt{d^2(u_1, v_1) + d^2(u_2, v_2)}$$

for all $u = (u_1, u_2), v = (v_1, v_2) \in [\mathbb{LC}(A_1, A_2)]^2$. As a consequence, the fuzzy solution of the initial value problem (20) is given by

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} q_1(t)A \oplus r_1(t)\tilde{A} \\ q_2(t)A \oplus r_2(t)\tilde{A} \end{bmatrix} = \begin{bmatrix} e^{-t}A \oplus \sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t)\tilde{A} \\ -\sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t)A \oplus e^{-t}\tilde{A} \end{bmatrix}.$$

Let $A = \frac{1}{2}A_1 \oplus \frac{1}{4}A_2 = (\frac{1}{4}; 2; 3)$ and $\tilde{A} = \frac{1}{4}A_1 \oplus \frac{1}{4}A_2 = (0; \frac{5}{4}; 2)$. Then, the graphical representation of the solution $x(t)$ is given in Fig. 1.

Case 2: Let $A_1 = (1; 3; 4)$ and $A_2 = (2; 6; 8)$ be two non-symmetric triangular fuzzy numbers. Note that since $A_1 \oplus (-\frac{1}{2})A_2 = A = (-3; 0; 3)$ is a symmetric fuzzy number with the symmetry point $x^* = 0$, it implies that the pair $\{A_1, A_2\}$ is not SLI. Then, the space $\mathbf{X}_1$ is given by

$$\mathbf{X}_1 = \{A = qA_1 \hat{\oplus} rA_2 \in \mathbb{LC}(A_1, A_2) : q, r \geq 0, \psi^{-1}(A) = \{(q, r), (0, 2ql_{1,2} + r)\}\},$$

where $l_{1,2} = \frac{1}{3}$ and the quotient space $\mathbb{R}^2 / \equiv_{\mathbb{LC}}$ is given by

$$\mathbb{R}^2 / \equiv_{\mathbb{LC}} := \left\{ [(q, r)]_{\equiv_{\mathbb{LC}}} = \left\{ (q, r), (0, \frac{2}{3}q + r) \right\} : (q, r) \in \mathbb{R}^2 \right\}.$$

Assume that $x_1(t), x_2(t)$ which take values in the space $\mathbf{X}_1$ are solutions of the initial value problem (17)-(18). Thus, there exist non-negative differentiable functions $\tilde{q}_i(t), \tilde{r}_i(t)$ such that the solution $x_i(t)$ can be represented by $x_i(t) = \tilde{q}_i(t)A \hat{\oplus} \tilde{r}_i(t)\tilde{A}$ and

$$\psi^{-1}(x(t)) = \{(\tilde{q}_i(t), \tilde{r}_i(t)), (0, 2\tilde{q}_i(t)l_{1,2} + \tilde{r}_i(t))\}$$

for all $t \in J$ and $i = 1, 2$. As a consequence, the fuzzy solution of the initial value problem (20) is given by

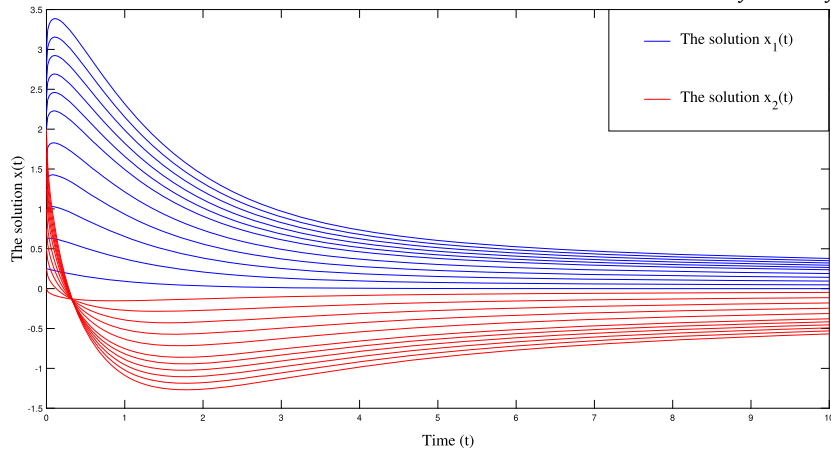


Fig. 1. The fuzzy solution $\mathbf{x}(t) = \mathbb{E}_{\frac{1}{2}}(Bt^{\frac{1}{2}})\mathbf{x}_0$ on the interval $[0, 10]$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

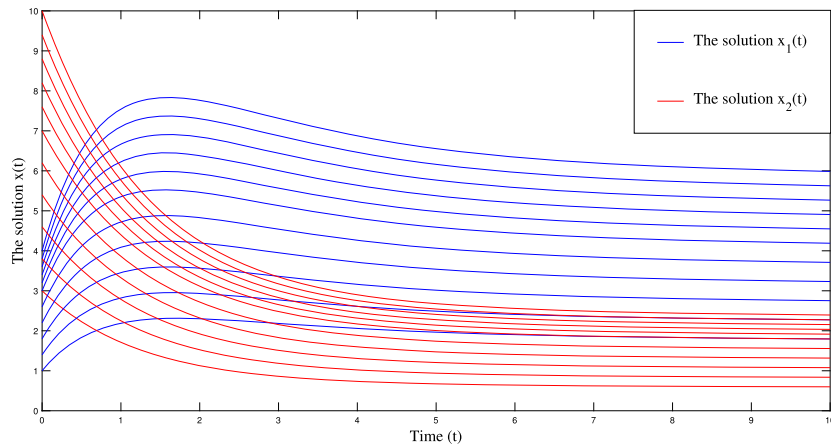


Fig. 2. The fuzzy solution $\mathbf{x}(t) = \mathbb{E}_{\frac{1}{2}}(Bt^{\frac{1}{2}})\mathbf{x}_0$ on the interval $[0, 10]$.

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \tilde{q}_1(t)A \hat{\oplus} \tilde{r}_1(t)\tilde{A} \\ \tilde{q}_2(t)A \hat{\oplus} \tilde{r}_2(t)\tilde{A} \end{bmatrix} = \begin{bmatrix} e^{-t}A \hat{\oplus} \sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t)\tilde{A} \\ -\sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t)A \hat{\oplus} e^{-t}\tilde{A} \end{bmatrix}.$$

Let $A = \frac{1}{2}A_1 \hat{\oplus} \frac{1}{4}A_2 = (1; 3; 4)$ and $\tilde{A} = A_1 \hat{\oplus} A_2 = (3; 7; 10)$. Then, the graphical representation of the solution $\mathbf{x}(t)$ is given in Fig. 2.

Furthermore, we have

$$\left[\mathbb{E}_{\frac{1}{2}}(Bt^{\frac{1}{2}}) \right]^{-1} = \frac{1}{e^{-2t} + t \left[\mathbb{E}_{1, \frac{3}{2}}(-t) \right]^2} \begin{bmatrix} e^{-t} & \sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t) \\ -\sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t) & e^{-t} \end{bmatrix}.$$

In addition, for each $t \in [0, 1]$, we have

$$\left\| \left[\mathbb{E}_{\frac{1}{2}}(Bt^{\frac{1}{2}}) \right]^{-1} \right\|_{\infty} = \frac{e^{-t} + \sqrt{t}\mathbb{E}_{1, \frac{3}{2}}(-t)}{e^{-2t} + t \left[\mathbb{E}_{1, \frac{3}{2}}(-t) \right]^2}$$

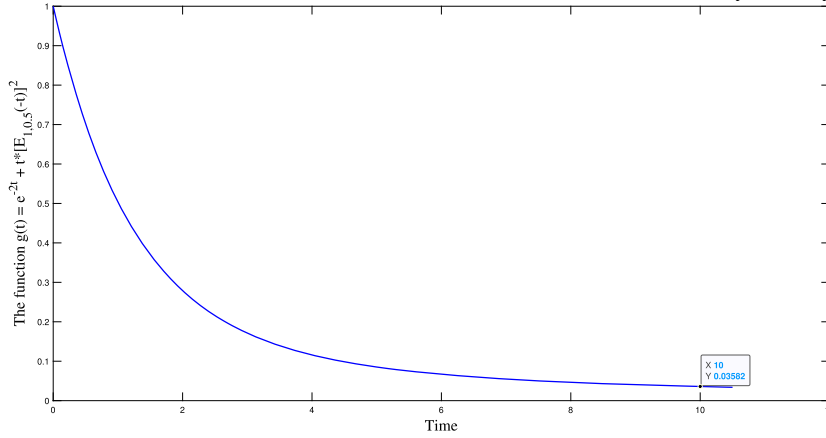


Fig. 3. The plot of the function $g_0(t) = e^{-2t} + t \left[E_{1, \frac{3}{2}}(-t) \right]^2$ on the interval $[0, 1]$.

$$\begin{aligned} &\leq \frac{\sqrt{2}}{\sqrt{e^{-2t} + t \left[E_{1, \frac{3}{2}}(-t) \right]^2}} \\ &\leq \frac{\sqrt{2}}{\min_{[0, 10]} \sqrt{e^{-2t} + t \left[E_{1, \frac{3}{2}}(-t) \right]^2}}. \end{aligned}$$

The function $g_0(t) = e^{-2t} + t \left[E_{1, \frac{3}{2}}(-t) \right]^2$ has a minimum $\min_{[0, 10]} g_0(t) = 0.03582$ whose graphical representation is given in Fig. 3.

Hence, it implies that $\left\| \left[E_{\frac{1}{2}}(Bt^{\frac{1}{2}}) \right]^{-1} \right\|_{\infty} \leq 39.481$. Next, the observability Gramian matrix for the system (17)-(19) can be calculated as follows:

$$\begin{aligned} W_0(t_f) &= \int_0^{t_f} E_{\frac{1}{2}}(B^T t^{\frac{1}{2}}) H^T H E_{\frac{1}{2}}(Bt^{\frac{1}{2}}) dt \\ &= \int_0^{t_f} \begin{bmatrix} e^{-2t} & e^{-t} \sqrt{t} E_{1, \frac{3}{2}}(-t) \\ e^{-t} \sqrt{t} E_{1, \frac{3}{2}}(-t) & t \left[E_{1, \frac{3}{2}}(-t) \right]^2 \end{bmatrix} dt, \end{aligned}$$

for each $t_f \in (0, 10]$. Thus, we have

$$\det(W_0(t_f)) = \left(\int_0^{t_f} e^{-2t} dt \right) \left(\int_0^{t_f} t \left[E_{1, \frac{3}{2}}(-t) \right]^2 dt \right) - \left(\int_0^{t_f} e^{-t} \sqrt{t} E_{1, \frac{3}{2}}(-t) dt \right)^2.$$

Next, consider two real-valued functions $g_1(t) = e^{-t}$ and $g_2(t) = \sqrt{t} E_{1, \frac{3}{2}}(-t)$ on the interval $[0, t_f]$. By using Cauchy-Schwarz-Buniakowsky inequality for integrals (see [40, Formula (12.311)]), we directly get that

$$\det(W_0(t_f)) = \left(\int_0^{t_f} g_1^2(t) dt \right) \left(\int_0^{t_f} g_2^2(t) dt \right) - \left(\int_0^{t_f} g_1(t) g_2(t) dt \right)^2 \geq 0. \quad (21)$$

It should be noted that the equality in (21) only holds if there exists an element $\kappa \in \mathbb{R}$ such that $g_2(t) = \kappa g_1(t)$ for all $t \in [0, t_f]$. However, it is obvious that there is no $\kappa \in \mathbb{R}$ such that $\sqrt{t} E_{1, \frac{3}{2}}(-t) = \kappa e^{-t}$ for all $t \in [0, t_f]$. It follows that the observability Gramian matrix $W_0(t_f)$ is non-singular for all $t_f \in (0, 10]$. Finally, we can conclude that the system (17)-(19) is completely observable on J .

As mentioned above, we have demonstrated the observability for the problem (17)–(19) in the LC space with A_1, A_2 not SLI, specifically when A_1 and A_2 are non-symmetric and $A_1 = kA_2$. Since the properties related to the case where A_1, A_2 are not SLI are quite complex (Section 2), the results for the remaining scenarios will be the focus of our future research.

5. Conclusions

The article studies the observability of fuzzy fractional evolution equations, that take values in the space $\mathbb{LC}(A_1, A_2)$ under Fréchet differentiability. The structural properties of the space $\mathbb{LC}(A_1, A_2)$ have been taken into consideration, especially in the case when $\{A_1, A_2\}$ is not SLI. Indeed, we provide a system of detailed conditions to ensure that the space $\mathbb{LC}(A_1, A_2)$ becomes a complete metric space. Furthermore, we present new concepts of differentiability and fractional calculus for the space of fuzzy-valued functions taking values in $\mathbb{LC}(A_1, A_2)$. This concept plays an important role in the study of fuzzy dynamical systems. Based on the foundation of this work, we are going to study some other qualitative properties of fractional dynamical systems in the space $\mathbb{LC}(A_1, A_2)$, such as stability, stabilization problems, and related control problems. Moreover, the study of nonlinear fractional dynamical systems in the space $\mathbb{LC}(A_1, A_2)$ remains an open question for future investigation.

CRedit authorship contribution statement

Nguyen Thi Thu Huyen: Writing – original draft, Investigation. **Nguyen Thi Kim Son:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Hoang Thi Phuong Thao:** Writing – original draft. **Nguyen Phuong Dong:** Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] P.R. Agarwal, V. Lakshmikantham, J.J. Nieto, On the concept of solution for fractional differential equations with uncertainty, *Nonlinear Anal.* 72 (2010) 2859–2862.
- [2] T. Allahviranloo, Z. Gouyandeh, A. Armand, Fuzzy fractional differential equations under generalized fuzzy Caputo derivative, *J. Intell. Fuzzy Syst.* 26 (2014) 1481–1490.
- [3] T. Allahviranloo, S. Salahshour, S. Abbasbandy, Explicit solutions of fractional differential equations with uncertainty, *Soft Comput.* 16 (2012) 297–302.
- [4] T.V. An, N.D. Phu, N.V. Hoa, A survey on non-instantaneous impulsive fuzzy differential equations involving the generalized Caputo fractional derivative in the short memory case, *Fuzzy Sets Syst.* 443 (2022) 160–197.
- [5] T.V. An, N.D. Phu, N.V. Hoa, The stabilization of uncertain dynamic systems involving the generalized Riemann-Liouville fractional derivative via linear state feedback control, *Fuzzy Sets Syst.* 472 (2023) 108697.
- [6] S. Arshad, V. Lupulescu, On the fractional differential equations with uncertainty, *Nonlinear Anal.* 74 (2011) 85–93.
- [7] M. Biglarbegian, A. Sadeghian, W. Melek, On the accessibility/controllability of fuzzy control systems, *Inf. Sci.* 202 (2012) 58–72.
- [8] B. Bede, *Mathematics of Fuzzy Sets and Fuzzy Logic*, Springer-Verlag, Berlin, 2013.
- [9] Y. Chalco-Cano, H. Roman-Flores, M.D. Jiménez-Gamero, Generalized derivative and π derivative for set-valued functions, *Inf. Sci.* 181 (2011) 2177–2188.
- [10] C. Dineshkumar, J.H. Jeong, Y.H. Joo, Observer-based fuzzy control for fractional order PMSG wind turbine systems with adaptive quantized-mechanism, *Commun. Nonlinear Sci. Numer. Simul.* 136 (2024) 108087.
- [11] Z. Ding, A. Kandel, On the observability of fuzzy dynamical control systems (II), *Fuzzy Sets Syst.* 115 (2) (2000) 261–277.
- [12] Z. Ding, M. Ma, A. Kandel, On the observability of fuzzy dynamical control systems (I), *Fuzzy Sets Syst.* 111 (2) (2000) 225–236.
- [13] N.P. Dong, H.V. Long, A. Khashtan, Optimal control of a fractional order model for granular SEIR epidemic with uncertainty, *Commun. Nonlinear Sci. Numer. Simul.* 88 (2020) 105312.
- [14] N.P. Dong, N.T.K. Son, T. Allahviranloo, H.T.T. Tam, Finite-time stability of mild solution to time-delay fuzzy fractional differential systems under granular computing, *Granul. Comput.* 8 (2) (2023) 223–239.
- [15] E. Esmi, L.C. de Barros, F.S. Pedro, B. Laiate, Banach spaces generated by strongly linearly independent fuzzy numbers, *Fuzzy Sets Syst.* 417 (2021) 110–129.
- [16] E. Esmi, B. Laiate, F.S. Pedro, L.C. Barros, Calculus for fuzzy functions with strongly linearly independent fuzzy coefficients, *Fuzzy Sets Syst.* 436 (2022) 1–31.
- [17] E. Esmi, F.S. Pedro, L.C. Barros, Fréchet derivative for linearly correlated fuzzy-valued function, *Inf. Sci.* 435 (2018) 150–160.
- [18] G.N. Galanis, Differentiability on semilinear spaces, *Nonlinear Anal.* 71 (2009) 4732–4738.
- [19] R. Goetschel, W. Voxman, Elementary fuzzy calculus, *Fuzzy Sets Syst.* 18 (1986) 31–43.
- [20] O. Kaleva, Fuzzy differential equations, *Fuzzy Sets Syst.* 24 (1987) 301–317.
- [21] A.A. Kilbas, H.M. Srivastava, J.J. Trujillo, *Theory and Applications of Fractional Differential Equations*, North-Holland Mathematics Studies, vol. 204, 2006.
- [22] S.R. Kou, D.L. Elliott, T.J. Tarn, Observability of nonlinear systems, *Inf. Control* 22 (1) (1973) 89–99.
- [23] V. Lupulescu, Hukuhara differentiability of interval-valued functions and interval differential equations on time scales, *Inf. Sci.* 248 (2013) 50–67.
- [24] V. Lupulescu, Fractional calculus for interval-valued functions, *Fuzzy Sets Syst.* 265 (2015) 63–85.
- [25] F.S. Pedro, E. Esmi, L.C. Barros, Calculus for linearly correlated fuzzy function using Fréchet derivative and Riemann integral, *Inf. Sci.* 512 (2020) 219–237.
- [26] F.S. Pedro, S.A.B. Salgado, D.E. Sanchez, E. Esmi, L.C. Barros, On fuzzy Laplace transform in linearly correlated fuzzy space, *Soft Comput.* 27 (2023) 1425–1438.
- [27] M.L. Puri, D.A. Ralescu, Differentials of fuzzy functions, *J. Math. Anal. Appl.* 91 (1983) 552–558.
- [28] I. Podlubny, *Fractional Differential Equations*, Mathematics in Science and Engineering, vol. 198, Technical University of Kosice, Slovak Republic, 1999, pp. 1–340.

- [29] S. Salahshour, T. Allahviranloo, S. Abbasbandy, Solving fuzzy fractional differential equations by fuzzy Laplace transforms, *Commun. Nonlinear Sci. Numer. Simul.* 17 (2012) 1372–1381.
- [30] S. Seikkala, On the fuzzy initial value problem, *Fuzzy Sets Syst.* 24 (1987) 319–330.
- [31] Y. Shen, First-order linear fuzzy differential equations on the space of linearly correlated fuzzy numbers, *Fuzzy Sets Syst.* 429 (2022) 136–168.
- [32] Y. Shen, Calculus for linearly correlated fuzzy number-valued functions, *Fuzzy Sets Syst.* 429 (2022) 101–135.
- [33] N.T.K. Son, Linear correlated fuzzy solutions of nonlocal problems for multi-order fractional differential systems, *Fuzzy Sets Syst.* 473 (2023) 108723.
- [34] N.T.K. Son, N.P. Dong, L.H. Son, A. Khastan, H.V. Long, Complete controllability for a class of fractional evolution equations with uncertainty, *Evol. Equ. Control Theory* 11 (2022).
- [35] N.T.K. Son, H.T.P. Thao, N.P. Dong, H.V. Long, Fractional calculus of linear correlated fuzzy-valued functions related to Fréchet differentiability, *Fuzzy Sets Syst.* 419 (2021) 35–66.
- [36] E.D. Sontag, *Mathematical Control Theory: Deterministic Finite Dimensional Systems*, 2nd ed., Springer, New York, NY, 2013.
- [37] L. Stefanini, B. Bede, Generalized Hukuhara differentiability of interval-valued functions and interval differential equations, *Nonlinear Anal.* 71 (2009) 1311–1328.
- [38] L. Stefanini, A generalization of Hukuhara difference and division for interval and fuzzy arithmetic, *Fuzzy Sets Syst.* 161 (2010) 1564–1584.
- [39] J. Zabczyk, *Mathematical Control Theory: An Introduction*, 2nd ed., Birkhäuser, Cham, 2020.
- [40] D. Zwillinger, *Table of Integrals, Series, and Products*, 7th ed., Academic Press, 2007.